# MATH 547 Course Notes

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**Abstract**

These are supplementary notes for Math 547 at USC for the Fall 2026 version of the course.

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# 1 Statistical learning as risk minimization

A supervised-learning problem begins with a domain set $\mathcal{X}$, a label set $\mathcal{Y}$, and the example space $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$. An example is a random pair $Z = (X, Y)$ with unknown distribution $D$ on $\mathcal{Z}$. The learner observes an independent and identically distributed sample

$$S = (Z_1, \ldots, Z_m) = ((X_1, Y_1), \ldots, (X_m, Y_m)) \sim D^m.$$

A hypothesis, predictor, or prediction rule is a function $h : \mathcal{X} \to \mathcal{Y}$, and a hypothesis class $\mathcal{H}$ is the collection of predictors available to the learner. Thus $D$ describes the population, $S$ is the observed data, and $\mathcal{H}$ describes the learner's admissible decisions.

**Example 1** (Papaya classification)**.** Represent a papaya by $x = (\text{color}, \text{softness}) \in \mathbb{R}^2$ and let $y \in \{0, 1\}$ indicate whether it tastes good. One possible hypothesis class is the class of indicator functions over axis-aligned rectangles: $h_R(x) = 1$ when $x \in R$ and $h_R(x) = 0$ otherwise and $R = \{x \in \mathbb{R}^2 : a \le x_1 \le b, \text{ and } c \le x_2 \le d\}$ for some valid choice of real numbers $a, b, c, d$. The distribution $D$ describes both which papayas occur and how taste depends on the recorded features.

The quality of a prediction rule is expressed by a loss function

$$\ell : \mathcal{H} \times \mathcal{Z} \longrightarrow \mathbb{R}_+,$$

where $\ell(h, z)$ is the cost of using $h$ on the example $z$. Two standard losses are

$$\ell_{0\text{-}1}(h, (x, y)) = \mathbf{1}\{h(x) \neq y\}, \qquad \ell_{\text{sq}}(h, (x, y)) = (h(x) - y)^2.$$

The first is used for classification and the second for regression.

We use Shalev-Shwartz and Ben-David's notation for the *risk* and *empirical risk* of $h$:

$$L_D(h) = \mathbb{E}_{Z \sim D}[\ell(h, Z)], \qquad L_S(h) = \frac{1}{m} \sum_{i=1}^{m} \ell(h, Z_i).$$

For 0–1 loss, the true and empirical risks have the following forms:

$$L_D(h) = \mathbb{P}_{(X,Y) \sim D}(h(X) \neq Y), \qquad L_S(h) = \frac{\#\{i : h(X_i) \neq Y_i\}}{m}.$$

In empirical-process notation, if a function $f$ represents the loss of a prediction rule, then the same comparison is written as $Pf$ versus $P_m f$, where $P$ is the unknown distribution and $P_m$ is the empirical measure.

The empirical risk minimization rule over $\mathcal{H}$ returns any

$$\widehat{h}_S \in \arg\min_{h \in \mathcal{H}} L_S(h).$$

When several hypotheses have the same minimum empirical risk, an ERM rule may return any one of them.

The restriction to $\mathcal{H}$ is essential. If every function is allowed, a predictor may simply memorize the sample. On an infinite domain, for example, one can reproduce every observed label and predict 0 elsewhere. Such a rule has zero empirical error, but its true error may be large. Choosing $\mathcal{H}$ before observing the sample is therefore an *inductive bias*. A restricted class can protect against overfitting, although it may exclude useful prediction rules; a richer class offers more candidates but is harder to search reliably with finite data.

For a distribution $\mathcal{D}$ over $\mathcal{X}$ and a labeling function $f : \mathcal{X} \to \{0, 1\}$, the notation $L_{(\mathcal{D}, f)}(h)$ denotes the 0–1 risk of $h$.

**Definition 1** (The Realizability Assumption)**.** There exists $h^\star \in \mathcal{H}$ such that

$$L_{(\mathcal{D},f)}(h^\star) = 0.$$

Note that this assumption implies that with probability 1 over random samples $S$, where the instances of $S$ are sampled according to $\mathcal{D}$ and are labeled by $f$, we have $L_S(h^\star) = 0$.

Under 0–1 loss, a realizable $h^*$ predicts correctly with probability one. Consequently, if $S \sim D^m$, then $L_S(h^*) = 0$ almost surely and every ERM satisfies $L_S(\widehat{h}_S) = 0$.

**Definition 2** (PAC Learnability)**.** A hypothesis class $\mathcal{H}$ is PAC learnable if there exist a function $m_{\mathcal{H}} : (0,1)^2 \to \mathbb{N}$ and a learning algorithm with the following property: For every $\epsilon, \delta \in (0,1)$, for every distribution $\mathcal{D}$ over $\mathcal{X}$, and for every labeling function $f : \mathcal{X} \to \{0,1\}$, if the realizable assumption holds with respect to $\mathcal{H}, \mathcal{D}, f$, then when running the learning algorithm on $m \geq m_{\mathcal{H}}(\epsilon, \delta)$ i.i.d. examples generated by $\mathcal{D}$ and labeled by $f$, the algorithm returns a hypothesis $h$ such that, with probability of at least $1 - \delta$ (over the choice of the examples),

$$L_{(\mathcal{D},f)}(h) \leq \epsilon.$$

Realizability may fail because of noise, unmeasured variables, or a misspecified hypothesis class. The appropriate benchmark is then the best risk achievable within $\mathcal{H}$.

**Definition 3** (Agnostic PAC Learnability for General Loss Functions)**.** A hypothesis class $\mathcal{H}$ is agnostic PAC learnable with respect to a set $Z$ and a loss function

$$\ell : \mathcal{H} \times Z \to \mathbb{R}_+$$

if there exist a function $m_{\mathcal{H}} : (0,1)^2 \to \mathbb{N}$ and a learning algorithm with the following property: For every $\epsilon, \delta \in (0,1)$ and for every distribution $\mathcal{D}$ over $Z$, when running the learning algorithm on $m \geq m_{\mathcal{H}}(\epsilon, \delta)$ i.i.d. examples generated by $\mathcal{D}$, the algorithm returns $h \in \mathcal{H}$ such that, with probability of at least $1 - \delta$ (over the choice of the $m$ training examples),

$$L_{\mathcal{D}}(h) \leq \min_{h' \in \mathcal{H}} L_{\mathcal{D}}(h') + \epsilon,$$

where

$$L_{\mathcal{D}}(h) = \mathbb{E}_{z \sim \mathcal{D}}[\ell(h, z)].$$

Koltchinskii defines excess risk as follows:

**Definition 4.** Let

$$\mathcal{E}(f) := \mathcal{E}_P(f) := \mathcal{E}_P(\mathcal{F}; f) := Pf - \inf_{g \in \mathcal{F}} Pg.$$

This quantity will be called the *excess risk* of $f \in \mathcal{F}$.

If the best rectangle classifier in the papaya example has risk 0.18, no learner restricted to rectangles can promise risk below 0.18. An agnostic guarantee with $\epsilon = 0.03$ instead asks for risk at most 0.21, with probability at least $1 - \delta$.

## 2 Finite hypothesis classes

Finite hypothesis classes exhibit two basic arguments used in learning theory. In the realizable setting, a bad hypothesis must escape detection on every observation. In the agnostic setting, empirical risk should approximate its population counterpart uniformly.

Assume first that $\mathcal{H}$ is a finite binary class, $S \sim D^m$ is i.i.d., the problem is realizable, and $\widehat{h}_S$ is an ERM for 0–1 loss. Fix $\epsilon > 0$ and define

$$\mathcal{H}_{\mathrm{B}} = \{h \in \mathcal{H} : L_D(h) > \epsilon\}.$$

For each $h \in \mathcal{H}_{\mathrm{B}}$, let

$$M_h = \{S : L_S(h) = 0\}, \qquad M = \bigcup_{h \in \mathcal{H}_{\mathrm{B}}} M_h.$$

The event $M_h$ says that a bad hypothesis nevertheless fits every observation.

**Lemma 1** (Failure containment)**.** *Under realizability,*

$$\{S : L_D(\widehat{h}_S) > \epsilon\} \subseteq M.$$

*Proof.* If $L_D(\widehat{h}_S) > \epsilon$, then $\widehat{h}_S \in \mathcal{H}_{\mathrm{B}}$. Realizability supplies a hypothesis with zero empirical error, so the ERM also satisfies $L_S(\widehat{h}_S) = 0$. Hence $\widehat{h}_S$ itself witnesses that $S \in M$. □

For a fixed $h \in \mathcal{H}_{\mathrm{B}}$, write $p = L_D(h)$. A fresh observation is classified correctly by $h$ with probability $1 - p < 1 - \epsilon$. Independence of the sample gives

$$\begin{aligned}
\mathbb{P}(M_h) &= \mathbb{P}\big(h(X_i) = Y_i \text{ for every } i = 1, \dots, m\big) \\
&= (1 - p)^m \\
&\le (1 - \epsilon)^m \\
&\le e^{-m\epsilon}.
\end{aligned}$$

Since the events $M_h$ need not be independent one bounds $\mathbb{P}(M)$ using a naive union bound, giving

$$\begin{aligned}
\mathbb{P}\big(L_D(\widehat{h}_S) > \epsilon\big) &\le \mathbb{P}(M) \\
&\le \sum_{h \in \mathcal{H}_{\mathrm{B}}} \mathbb{P}(M_h) \\
&\le |\mathcal{H}| e^{-m\epsilon}.
\end{aligned}$$

**Theorem 1** (Finite realizable classes)**.** *Let $\mathcal{H}$ be a finite binary hypothesis class, let $\delta \in (0, 1)$ and $\epsilon > 0$, and assume realizability. If*

$$m \ge \frac{\log|\mathcal{H}| + \log(1/\delta)}{\epsilon} = \frac{\log(|\mathcal{H}|/\delta)}{\epsilon},$$

*then every ERM satisfies*

$$\mathbb{P}_{S \sim D^m}\big(L_D(\widehat{h}_S) \le \epsilon\big) \ge 1 - \delta.$$

*Thus every finite class is PAC learnable by ERM.*

*Proof.* The sample-size condition implies $e^{-m\epsilon} \le \delta/|\mathcal{H}|$. Substitution into the preceding failure bound gives

$$\mathbb{P}(L_D(\widehat{h}_S) > \epsilon) \le |\mathcal{H}| e^{-m\epsilon} \le \delta.$$

□

Quick example: if $|\mathcal{H}| = 1000$, $\epsilon = 0.10$, and $\delta = 0.05$, then

$$m \geq \frac{\log(1000/0.05)}{0.10} \approx 99.04,$$

so $m = 100$ is sufficient to learn a hypothesis with 95% confidence with 90% accurancy.

In the agnostic setting, an ERM need not have zero empirical error. The replacement is simultaneous approximation of every true risk by its empirical risk.

**Definition 5** ($\epsilon$-representative sample)**.** A training set $S$ is called $\epsilon$-representative (with respect to domain $Z$, hypothesis class $\mathcal{H}$, loss function $\ell$, and distribution $\mathcal{D}$) if

$$\forall h \in \mathcal{H}, \qquad |L_S(h) - L_{\mathcal{D}}(h)| \leq \epsilon.$$

**Definition 6** (Uniform Convergence)**.** We say that a hypothesis class $\mathcal{H}$ has the uniform convergence property (with respect to a domain $Z$ and a loss function $\ell$) if there exists a function

$$m_{\mathcal{H}}^{\mathrm{UC}} : (0,1)^2 \to \mathbb{N}$$

such that for every $\epsilon, \delta \in (0,1)$ and for every probability distribution $\mathcal{D}$ over $Z$, if $S$ is a sample of $m \geq m_{\mathcal{H}}^{\mathrm{UC}}(\epsilon, \delta)$ examples drawn i.i.d. according to $\mathcal{D}$, then, with probability of at least $1 - \delta$, $S$ is $\epsilon$-representative.

Uniformity is necessary because $\widehat{h}_S$ is chosen after the sample has been observed.

**Lemma 2** (ERM transfer)**.** *Suppose a true-risk minimizer $h^* \in \arg\min_{h \in \mathcal{H}} L_D(h)$ exists. If $S$ is $(\epsilon/2)$-representative, then every ERM satisfies*

$$L_D(\widehat{h}_S) \leq L_D(h^*) + \epsilon.$$

*Proof.* Representativeness, empirical optimality, and representativeness once more give

$$\begin{aligned} L_D(\widehat{h}_S) &\leq L_S(\widehat{h}_S) + \frac{\epsilon}{2} \\ &\leq L_S(h^*) + \frac{\epsilon}{2} \\ &\leq L_D(h^*) + \epsilon. \end{aligned}$$

□

**Lemma 3** (Hoeffding's inequality)**.** *Let $\theta_1, \ldots, \theta_m$ be a sequence of i.i.d. random variables and assume that for all $i$,*

$$\mathbb{E}[\theta_i] = \mu \qquad \textit{and} \qquad \mathbb{P}[a \leq \theta_i \leq b] = 1.$$

*Then, for any $\epsilon > 0$,*

$$\mathbb{P}\left[\left|\frac{1}{m}\sum_{i=1}^{m} \theta_i - \mu\right| > \epsilon\right] \leq 2\exp\left(-\frac{2m\epsilon^2}{(b-a)^2}\right).$$

For a fixed $h$ and a loss in $[0,1]$, take $\theta_i = \ell(h, Z_i)$ and apply Hoeffding's inequality with $\epsilon = \alpha$. Then

$$\mathbb{P}(|L_S(h) - L_D(h)| > \alpha) \leq 2e^{-2m\alpha^2}.$$

A union bound over the finite class now gives

$$\mathbb{P}\left(\sup_{h \in \mathcal{H}} |L_S(h) - L_D(h)| > \alpha\right) \leq 2|\mathcal{H}|e^{-2m\alpha^2}.$$

Consequently,

$$m \geq \frac{\log(2|\mathcal{H}|/\delta)}{2\alpha^2}$$

is sufficient for an $\alpha$-representative sample with probability at least $1-\delta$.

**Theorem 2** (Finite-class agnostic ERM)**.** *Let $\mathcal{H}$ be finite, let $\ell : \mathcal{H} \times \mathcal{Z} \to [0,1]$, and let $\epsilon, \delta \in (0,1)$. If*

$$m \geq \frac{2\log(2|\mathcal{H}|/\delta)}{\epsilon^2},$$

*then, with probability at least $1-\delta$,*

$$L_D(\widehat{h}_S) \leq \min_{h \in \mathcal{H}} L_D(h) + \epsilon.$$

*Consequently, $\mathcal{H}$ is agnostically PAC learnable by ERM.*

The realizable result scales as $1/\epsilon$, whereas the agnostic result scales as $1/\epsilon^2$.

# 3 No-Free-Lunch and inductive bias

Finite-class bounds show that restriction can make learning possible. The No-Free-Lunch principle explains why some restriction is unavoidable. Consider binary classification on $\mathcal{X}$ with 0–1 loss. A learner $A$ receives $S \sim D^m$ and returns $A(S) : \mathcal{X} \to \{0,1\}$. If arbitrary target functions are permitted, labels on points absent from the sample contain no information for the learner.

**Theorem 3** (No-Free-Lunch)**.** *Let $A$ be any learning algorithm for binary classification with 0–1 loss. If $m < |\mathcal{X}|/2$, then there is a distribution $D$ on $\mathcal{X} \times \{0,1\}$ such that*

1. *some function $f : \mathcal{X} \to \{0,1\}$ satisfies $L_D(f) = 0$; and*
2. *with probability at least $1/7$ over $S \sim D^m$,*

$$L_D(A(S)) \geq \frac{1}{8}.$$

*Proof.* Choose a subset $C \subseteq \mathcal{X}$ with $|C| = 2m$. There are $T = 2^{2m}$ functions from $C$ to $\{0,1\}$; enumerate them as $f_1, \ldots, f_T$. For each $i \in [T] := \{1, \ldots, T\}$, define a distribution $D_i$ on $C \times \{0,1\}$ by

$$D_i(\{(x,y)\}) = \begin{cases} 1/|C|, & y = f_i(x), \\ 0, & \text{otherwise.} \end{cases}$$

Thus $D_i$ chooses $X$ uniformly from $C$ and sets $Y = f_i(X)$, so $L_{D_i}(f_i) = 0$.

We first prove that every learner receiving $m$ examples from $C \times \{0,1\}$ satisfies

$$\max_{i \in [T]} \mathbb{E}_{S \sim D_i^m}\left[L_{D_i}(A(S))\right] \geq \frac{1}{4}. \tag{1}$$

This will identify at least one realizable distribution on which the learner's expected risk is large.

There are $k = (2m)^m$ ordered sequences of $m$ instances from $C$; enumerate them as $S_1, \ldots, S_k$. If $S_j = (x_1, \ldots, x_m)$, let

$$S_j^i = \big((x_1, f_i(x_1)), \ldots, (x_m, f_i(x_m))\big)$$

be the corresponding sample labeled by $f_i$. Under $D_i^m$, the samples $S_1^i, \ldots, S_k^i$ are equally likely. Therefore

$$\mathbb{E}_{S\sim D_i^m}\left[L_{D_i}(A(S))\right] = \frac{1}{k}\sum_{j=1}^{k} L_{D_i}(A(S_j^i)). \tag{2}$$

The maximum over target labelings is at least their average, and the average over instance sequences is at least its smallest summand. Hence

$$\begin{aligned}\max_{i\in[T]} \frac{1}{k}\sum_{j=1}^{k} L_{D_i}(A(S_j^i)) &\geq \frac{1}{T}\sum_{i=1}^{T}\frac{1}{k}\sum_{j=1}^{k} L_{D_i}(A(S_j^i)) \\ &= \frac{1}{k}\sum_{j=1}^{k}\frac{1}{T}\sum_{i=1}^{T} L_{D_i}(A(S_j^i)) \\ &\geq \min_{j\in[k]} \frac{1}{T}\sum_{i=1}^{T} L_{D_i}(A(S_j^i)). \end{aligned} \tag{3}$$

Fix $j \in [k]$, write $S_j = (x_1, \ldots, x_m)$, and let $v_1, \ldots, v_p$ be the points of $C$ that do not occur in $S_j$. At most $m$ distinct points occur in the sequence, so $p \geq m$. For every $h : C \to \{0, 1\}$ and every $i$,

$$\begin{aligned} L_{D_i}(h) &= \frac{1}{2m}\sum_{x\in C} \mathbf{1}\{h(x) \neq f_i(x)\} \\ &\geq \frac{1}{2m}\sum_{r=1}^{p} \mathbf{1}\{h(v_r) \neq f_i(v_r)\} \\ &\geq \frac{1}{2p}\sum_{r=1}^{p} \mathbf{1}\{h(v_r) \neq f_i(v_r)\}. \end{aligned} \tag{4}$$

Applying this inequality to $h = A(S_j^i)$ and averaging over all target labelings gives

$$\begin{aligned}\frac{1}{T}\sum_{i=1}^{T} L_{D_i}(A(S_j^i)) &\geq \frac{1}{T}\sum_{i=1}^{T}\frac{1}{2p}\sum_{r=1}^{p} \mathbf{1}\{A(S_j^i)(v_r) \neq f_i(v_r)\} \\ &= \frac{1}{2p}\sum_{r=1}^{p}\frac{1}{T}\sum_{i=1}^{T} \mathbf{1}\{A(S_j^i)(v_r) \neq f_i(v_r)\} \\ &\geq \frac{1}{2}\min_{r\in[p]} \frac{1}{T}\sum_{i=1}^{T} \mathbf{1}\{A(S_j^i)(v_r) \neq f_i(v_r)\}. \end{aligned} \tag{5}$$

Fix $r \in [p]$. Pair the $T$ functions $f_1, \ldots, f_T$ so that the two functions in each pair agree at every point of $C$ except $v_r$. If $f_i$ and $f_{i'}$ form such a pair, then $v_r$ does not occur in $S_j$, and therefore $S_j^i = S_j^{i'}$. The learner consequently makes the same prediction at $v_r$ on the two samples. Since $f_i(v_r) \neq f_{i'}(v_r)$, exactly one of that pair's two error indicators equals 1:

$$\mathbf{1}\{A(S_j^i)(v_r) \neq f_i(v_r)\} + \mathbf{1}\{A(S_j^{i'})(v_r) \neq f_{i'}(v_r)\} = 1.$$

There are $T/2$ disjoint pairs, so

$$\frac{1}{T}\sum_{i=1}^{T} \mathbf{1}\{A(S_j^i)(v_r) \neq f_i(v_r)\} = \frac{1}{2}.$$

This equality holds for every $r \in [p]$. Substitution into (5) shows that

$$\frac{1}{T}\sum_{i=1}^{T} L_{D_i}(A(S_j^i)) \geq \frac{1}{4}$$

for every $j$. Combining this with (3) and (2) proves (1).

Choose $i^* \in [T]$ attaining the maximum in (1). Because $D_{i^*}$ is supported on $C$, extend $f_{i^*}$ arbitrarily from $C$ to $\mathcal{X}$ and regard the learner's output only through its restriction to $C$. With $D = D_{i^*}$ and this extension $f$, we have $L_D(f) = 0$ and

$$\mathbb{E}_{S\sim D^m}\left[L_D(A(S))\right] \geq \frac{1}{4}.$$

It remains to convert this expectation bound into the claimed probability. Let $R = L_D(A(S)) \in [0,1]$ and put $q = \mathbb{P}(R \geq 1/8)$. Splitting the expectation over this event and its complement yields

$$\mathbb{E}R \leq q \cdot 1 + (1-q)\frac{1}{8} = \frac{1}{8} + \frac{7}{8}q.$$

Since $\mathbb{E}R \geq 1/4$, it follows that $q \geq 1/7$. Thus, with probability at least $1/7$, the learner has risk at least $1/8$, as required. □

As a comment, if one take the distribution in the proof to be $D_I$, where $I$ is uniformly distributed on $[T]$, and argue directly via conditionining that this distribution gives the desired true-risk, then the resulting proof should be a a lot shorter.

In the statement of The No-Free Lunch theorem the order of quantifiers are essential: the result is $\forall A\, \exists D$, not $\exists D\, \forall A$. The hard distribution may depend on the learner. The theorem says that no learner succeeds uniformly over all possible binary tasks without encoding prior structure.

**Corollary 1.** *If $\mathcal{X}$ is infinite, the class of all functions from $\mathcal{X}$ to $\{0,1\}$ is not PAC learnable.*

*Proof.* If the class were PAC learnable, choose $\epsilon < 1/8$ and $\delta < 1/7$, and then take $m \geq m_{\mathcal{H}}(\epsilon, \delta)$. An infinite domain contains a set of size $2m$, so the theorem produces a realizable distribution on which the PAC guarantee fails. □

The choice of a hypothesis class introduces two distinct sources of error. If $\widehat{h}_S$ is an ERM and a true-risk minimizer exists in $\mathcal{H}$, then

$$L_D(\widehat{h}_S) = \underbrace{\min_{h\in\mathcal{H}} L_D(h)}_{\text{approximation error}} + \underbrace{\left[L_D(\widehat{h}_S) - \min_{h\in\mathcal{H}} L_D(h)\right]}_{\text{estimation error}}.$$

Approximation error is the cost of restricting the learner to $\mathcal{H}$. Enlarging $\mathcal{H}$ cannot increase this term. Estimation error is the excess risk incurred because empirical risk is only an estimate of true risk. Richer classes are generally harder to distinguish with a fixed amount of data and therefore require more samples to differentiate.

For nested classes $\mathcal{H}_1 \subseteq \mathcal{H}_2 \subseteq \cdots$, structural risk minimization makes this tradeoff explicit by selecting

$$\widehat{k} \in \arg\min_k \left\{ \min_{h\in\mathcal{H}_k} L_S(h) + \mathrm{pen}(k, m, \delta) \right\}.$$

The empirical-risk term favors expressive classes; the penalty accounts for their larger estimation error. Such a penalty must be justified by a complexity bound rather than chosen arbitrarily.

# 4 VC dimension

Cardinality is inadequate for many natural hypothesis classes because they contain infinitely many functions. VC dimension measures instead how freely the class can label finite subsets of its domain.

**Definition 7** (Restriction of $\mathcal{H}$ to $C$)**.** Let $\mathcal{H}$ be a class of functions from $\mathcal{X}$ to $\{0,1\}$ and let $C = \{c_1, \ldots, c_m\} \subset \mathcal{X}$. The restriction of $\mathcal{H}$ to $C$ is the set of functions from $C$ to $\{0,1\}$ that can be derived from $\mathcal{H}$. That is,

$$\mathcal{H}_C = \{(h(c_1), \ldots, h(c_m)) : h \in \mathcal{H}\},$$

where we represent each function from $C$ to $\{0,1\}$ as a vector in $\{0,1\}^{|C|}$.

**Definition 8** (Shattering)**.** A hypothesis class $\mathcal{H}$ shatters a finite set $C \subset \mathcal{X}$ if the restriction of $\mathcal{H}$ to $C$ is the set of all functions from $C$ to $\{0,1\}$. That is,

$$|\mathcal{H}_C| = 2^{|C|}.$$

**Definition 9** (VC-dimension)**.** The VC-dimension of a hypothesis class $\mathcal{H}$, denoted $\mathrm{VCdim}(\mathcal{H})$, is the maximal size of a set $C \subset \mathcal{X}$ that can be shattered by $\mathcal{H}$. If $\mathcal{H}$ can shatter sets of arbitrarily large size, we say that $\mathcal{H}$ has infinite VC-dimension.

Shattering requires every binary labeling of $C$ to be realized, generally by different hypotheses. To prove $\mathrm{VCdim}(\mathcal{H}) = d$, one must exhibit a shattered $d$-point set and prove that no set of $d+1$ points is shattered.

**Example 2** (Thresholds)**.** Let $\mathcal{H} = \{h_a : a \in \mathbb{R}\}$, where $h_a(x) = \mathbf{1}\{x < a\}$. A single point can receive either label by moving $a$ across it. For $c_1 < c_2$, however, the labeling $(0,1)$ is impossible. Hence $\mathrm{VCdim}(\mathcal{H}) = 1$.

**Example 3** (Intervals)**.** Let $h_{a,b}(x) = \mathbf{1}\{x \in (a,b)\}$. Two ordered points can be shattered. For three points $c_1 < c_2 < c_3$, the labeling $(1,0,1)$ cannot be realized by an interval. Hence the VC dimension is 2.

**Example 4** (Axis-aligned rectangles)**.** The indicators of axis-aligned rectangles in $\mathbb{R}^2$ have VC dimension 4. For the lower bound, place four points at the top, bottom, left, and right of a small diamond. Adjusting the four sides of a rectangle realizes every subset of these points. For the upper bound, take any five points and select a leftmost, rightmost, lowest, and highest point. These roles use at most four points, so some point $c$ is unselected. Every rectangle containing the four extreme points also contains $c$; thus the labeling that is 1 on the extreme points and 0 at $c$ is impossible.

**Lemma 4.** *If $\mathcal{H}$ is finite, then*

$$\mathrm{VCdim}(\mathcal{H}) \le \log_2 |\mathcal{H}|.$$

*Proof.* Restriction cannot create new hypotheses, so $|\mathcal{H}_C| \le |\mathcal{H}|$ for every finite $C$. If $C$ is shattered, then $2^{|C|} = |\mathcal{H}_C| \le |\mathcal{H}|$. Taking base-2 logarithms proves the claim. □

The inequality can be loose. Thresholds on the finite ordered domain $\{1, \ldots, k\}$ form a class whose cardinality grows with $k$, although its VC dimension remains 1. Conversely, VC dimension is not simply the number of real parameters in a formula: a single real parameter can encode increasingly oscillatory labeling patterns.

An important geometric example is the class of homogeneous halfspaces

$$\mathcal{H}_d = \{h_w : w \in \mathbb{R}^d\}, \qquad h_w(x) = \mathbf{1}\{\langle w, x\rangle > 0\}.$$

**Theorem 4.** $\mathrm{VCdim}(\mathcal{H}_d) = d$.

*Proof.* For the lower bound, take the standard basis $e_1, \ldots, e_d$. Given labels $y_i \in \{0, 1\}$, choose

$$w = (2y_1 - 1, \ldots, 2y_d - 1).$$

Then $\langle w, e_i \rangle = 2y_i - 1$, so $h_w(e_i) = y_i$. Thus the basis is shattered.

For the upper bound, take any $d + 1$ points $x_1, \ldots, x_{d+1}$. They are linearly dependent, so there are coefficients $\alpha_i$, not all zero, with $\sum_i \alpha_i x_i = 0$. Multiply all coefficients by $-1$ if necessary so that at least one $\alpha_i > 0$, and assign label 1 when $\alpha_i > 0$ and label 0 when $\alpha_i \leq 0$. If some $w$ realized these labels, then

$$0 = \left\langle w, \sum_i \alpha_i x_i \right\rangle = \sum_i \alpha_i \langle w, x_i \rangle > 0,$$

since every summand is nonnegative and those with $\alpha_i > 0$ are strictly positive. This contradiction gives a labeling that no homogeneous halfspace realizes. □

VC dimension also transfers the No-Free-Lunch obstruction to a fixed hypothesis class. If $\mathcal{H}$ shatters a set of size $2m$, every target labeling used in the adversarial construction agrees on that set with some member of $\mathcal{H}$. The resulting distributions are therefore realizable by $\mathcal{H}$.

**Theorem 5** (Infinite-VC obstruction)**.** *If* $\mathrm{VCdim}(\mathcal{H}) = \infty$*, then* $\mathcal{H}$ *is not PAC learnable.*

*Proof.* For every proposed sample size $m$, choose a $2m$-point set shattered by $\mathcal{H}$ and apply the No-Free-Lunch construction on that set. It produces a distribution realizable by $\mathcal{H}$ for which the learner has risk at least $1/8$ with probability at least $1/7$. □

Finite VC dimension is therefore necessary for binary PAC learnability. The fundamental theorem of statistical learning shows that it is also sufficient.

# 5 Growth functions

If a hypothesis class $\mathcal{H}$ has VC dimension $d$ then the largest subset of the domain $\mathcal{X}$ which it can shatter has size $d$. It turns out that this fact implies that the number of distinct labelings on a subset of size $m$ (i.e. the growth function) can grow fast as $m^d$. In this section we will prove this fact. We begin with a definition.

**Definition 10** (Growth Function)**.** Let $\mathcal{H}$ be a hypothesis class. Then the growth function of $\mathcal{H}$, denoted $\tau_{\mathcal{H}} : \mathbb{N} \to \mathbb{N}$, is defined as

$$\tau_{\mathcal{H}}(m) = \max_{C \subset \mathcal{X} : |C| = m} |\mathcal{H}_C|.$$

In words, $\tau_{\mathcal{H}}(m)$ is the number of different functions from a set $C$ of size $m$ to $\{0, 1\}$ that can be obtained by restricting $\mathcal{H}$ to $C$.

Clearly $\tau_{\mathcal{H}}(m) \leq 2^m$, and

$$\mathrm{VCdim}(\mathcal{H}) \geq m \quad \iff \quad \tau_{\mathcal{H}}(m) = 2^m.$$

Thus VC dimension is the largest scale at which $\mathcal{H}$ can realize every possible labeling.

**Example 5** (Threshold growth)**.** On $m$ ordered points, a threshold places its cut in one of $m+1$ gaps. Therefore

$$\tau_{\mathcal{H}}(m) = m + 1,$$

even though the class of thresholds on $\mathbb{R}$ is infinite.

**Example 6** (Interval growth)**.** On $m$ ordered points, a nonempty interval selects one consecutive block. Including the empty labeling gives

$$\tau_{\mathcal{H}}(m) = 1 + \frac{m(m+1)}{2}.$$

The quadratic growth is consistent with VC dimension 2.

We now show that $\tau_{\mathcal{H}}(m)$ grows polynomially. The proof reduces to the combinatorial fact that the growth function is comparable to a binomial coefficient.

**Theorem 6** (Sauer–Shelah–Perles)**.** *If* $\mathrm{VCdim}(\mathcal{H}) \le d < \infty$*, then for every* $m$*,*

$$\tau_{\mathcal{H}}(m) \le \sum_{i=0}^{d} \binom{m}{i}.$$

*In particular, if* $m > d+1$*, then*

$$\tau_{\mathcal{H}}(m) \le \left(\frac{em}{d}\right)^{d}.$$

*Proof.* We may assume that $m \ge d$ (otherwise all the bounds are trivial). We prove the bound by induction on $m+d$. If $m+d=2$ then $m=d=1$ and $\tau_H$ is exactly $2 = \binom{1}{0} + \binom{1}{1}$, proving the base case. Assume then that $m+d>2$. Define

$$T(m,d) = \sup\{|\mathcal{H}_C| : |C| = m,\ \mathrm{VCdim}(\mathcal{H}) \le d\}.$$

The base case implies that $T(1,1) = 2$. Let $\mathcal{H}$ and $C = \{c_1, \ldots, c_m\}$ be chosen so that $|\mathcal{H}_C| = T(m,d)$. Let $Y \subset \mathcal{H}_{C'}$ denote all classifiers on $C'$ that lift to exactly two classifiers in $\mathcal{H}_C$ (i.e. if $y \in Y$ then $(0,y)$ and $(1,y)$ are both in $\mathcal{H}_C$). Note that every function in $\mathcal{H}_C \setminus Y$ lifts to exactly one classifier in $\mathcal{H}_C$. Since $VCdim(\mathcal{H}_C) = VCdim(\mathcal{H}) = d$ the definition of $Y$ implies that $VCdim(Y) \le d-1$ (if $Y$ instead shattered a subset of $C'$ of size $d$ then its lift in $\mathcal{H}_C$ would shatter that subset and $c_1$, which collectively are a subset of size $d+1$, contradicting the VC bound). Thus the preceding decomposition yields

$$T(m,d) = |\mathcal{H}_C| = |\mathcal{H}_{C'} \setminus Y| + 2|Y| = |\mathcal{H}_{C'}| + |Y| \le T(m-1,d) + T(m-1,d-1).$$

Pascal's identity and the inductive hypothesis then give

$$\begin{aligned} T(m,d) &\le \sum_{i=0}^{d} \binom{m-1}{i} + \sum_{i=0}^{d-1} \binom{m-1}{i} \\ &= \sum_{i=0}^{d} \left[ \binom{m-1}{i} + \binom{m-1}{i-1} \right] \\ &= \sum_{i=0}^{d} \binom{m}{i}, \end{aligned}$$

where $\binom{m-1}{-1} = 0$. Finally, the standard binomial estimate

$$\sum_{i=0}^{d} \binom{m}{i} \leq \left(\frac{em}{d}\right)^d, \qquad m > d+1,$$

gives the stated polynomial bound. □

*Remark* 1 (VC index convention). For a nonempty VC class of sets $\mathcal{C}$, Giné and Nickl define the VC index $v(\mathcal{C})$ as the smallest $k$ for which no $k$-point set is shattered. Thus $v(\mathcal{C}) = \mathrm{VCdim}(\mathcal{C}) + 1$, and their binomial sum through $v - 1$ is the same bound as the sum through $d$ above.

The growth function gives an upper bound on the effective size of a (possibly infinite) hypothesis class over a finite sample. If $\mathrm{VCdim}(\mathcal{H}) = d < \infty$, then

$$\log \tau_{\mathcal{H}}(m) \leq d \log\left(\frac{em}{d}\right).$$

The effective log-cardinality therefore grows like $d \log m$, rather than linearly in $m$. Combined with symmetrization and concentration, this observation allows us to extend the finite-class arguments for proving learnability of infinite sized classes with finite VC dimension.

## 6 Uniform convergence and the Fundamental Theorem

The growth function permits the finite-class concentration argument to be used for hypothesis classes with finite VC-dimension. The main difference is how we view the difference between the true and empirical risks. In the finite-class we viewed it as the average of centered *random* sample losses. This is somewhat problematic in the finite VC-dimension case since using the growth function bound requires us to condition on the sample set, meaning the losses are no longer random. Instead we will view it as a difference of iid losses by pulling out the expectation, allowing us to replace the true risk with another average of sample losses coming from *ghost samples*. In particular the difference is now an average of iid differences of iid random variables. Therefore, after conditioning on the true and ghost samples, this becomes an average of iid bounded and symmetric random variables, for which the Hoeffding inequality will apply.

**Theorem 7** (Uniform convergence from effective size)**.** *Let $\mathcal{H}$ be a class and let $\tau_{\mathcal{H}}$ be its growth function. Then, for every distribution $D$ and every $\delta \in (0,1)$, with probability at least $1-\delta$ over the choice of $S \sim D^m$, we have*

$$\forall h \in \mathcal{H}, \qquad |L_D(h) - L_S(h)| \leq \frac{8 + 2\sqrt{\log(\tau_{\mathcal{H}}(2m))}}{\delta\sqrt{2m}}.$$

*Proof.* For a labeled example $z = (x, y)$, write $\ell(h, z) = \mathbf{1}\{h(x) \neq y\}$. Let $S = (Z_1, \ldots, Z_m) \sim D^m$ and define the nonnegative random variable

$$G(S) = \sup_{h \in \mathcal{H}} |L_D(h) - L_S(h)|.$$

We first prove the expectation bound

$$\mathbb{E}_S G(S) \leq \frac{8 + 2\sqrt{\log(\tau_{\mathcal{H}}(2m))}}{\sqrt{2m}}. \tag{6}$$

Let $S' = (Z'_1, \dots, Z'_m) \sim D^m$ be independent of $S$. For each $h \in \mathcal{H}$,

$$L_D(h) = \mathbb{E}_{S'} L_{S'}(h).$$

Consequently, for every fixed $S$, the triangle inequality for expectations and the fact that a supremum of expectations is at most the expectation of a supremum give

$$\begin{aligned} G(S) &= \sup_{h \in \mathcal{H}} \left| \mathbb{E}_{S'} \left[ L_{S'}(h) - L_S(h) \right] \right| \\ &\le \sup_{h \in \mathcal{H}} \mathbb{E}_{S'} \left| L_{S'}(h) - L_S(h) \right| \\ &\le \mathbb{E}_{S'} \sup_{h \in \mathcal{H}} \left| L_{S'}(h) - L_S(h) \right|. \end{aligned}$$

Taking expectation with respect to $S$ therefore yields

$$\begin{aligned} \mathbb{E}_S G(S) &\le \mathbb{E}_{S,S'} \sup_{h \in \mathcal{H}} \left| L_{S'}(h) - L_S(h) \right| \\ &= \mathbb{E}_{S,S'} \sup_{h \in \mathcal{H}} \left| \frac{1}{m} \sum_{i=1}^{m} \big( \ell(h, Z'_i) - \ell(h, Z_i) \big) \right|. \end{aligned} \tag{7}$$

We next randomize the signs of the summands. Fix $\sigma = (\sigma_1, \dots, \sigma_m) \in \{-1, 1\}^m$. For each index with $\sigma_i = -1$, interchange $Z_i$ and $Z'_i$. The $2m$ examples are i.i.d., so this interchange leaves the joint distribution of $(S, S')$ unchanged, while it changes the sign of the $i$th difference. Thus the last expression in (7) equals

$$\mathbb{E}_{S,S'} \sup_{h \in \mathcal{H}} \left| \frac{1}{m} \sum_{i=1}^{m} \sigma_i \big( \ell(h, Z'_i) - \ell(h, Z_i) \big) \right|$$

for every fixed sign vector $\sigma$. We may therefore average over independent uniform signs $\sigma_1, \dots, \sigma_m$ and exchange the order of expectation to obtain

$$\mathbb{E}_S G(S) \le \mathbb{E}_{S,S'} \mathbb{E}_\sigma \sup_{h \in \mathcal{H}} \left| \frac{1}{m} \sum_{i=1}^{m} \sigma_i \big( \ell(h, Z'_i) - \ell(h, Z_i) \big) \right|. \tag{8}$$

Fix the two samples, and let $C$ be the set of instances appearing among $Z_1, \dots, Z_m, Z'_1, \dots, Z'_m$. Two hypotheses that agree on $C$ have the same loss on all $2m$ labeled examples. Hence the supremum in (8) can be replaced by a maximum over the finite restriction

$$\mathcal{H}_C = \{ h|_C : h \in \mathcal{H} \}, \qquad |\mathcal{H}_C| \le \tau_{\mathcal{H}}(2m).$$

For each $g \in \mathcal{H}_C$, choose a representative $h_g \in \mathcal{H}$ and put

$$\theta_g = \frac{1}{m} \sum_{i=1}^{m} \sigma_i \big( \ell(h_g, Z'_i) - \ell(h_g, Z_i) \big).$$

Conditional on $S, S'$, the summands are independent, have mean zero, and take values in $[-1, 1]$, an interval of length 2. Hoeffding's inequality therefore gives, for every $\rho > 0$,

$$\mathbb{P}_\sigma \big( |\theta_g| > \rho \big) \le 2 \exp\left( -\frac{2m^2 \rho^2}{4m} \right) = 2 \exp\left( -\frac{m \rho^2}{2} \right).$$

A union bound over the restrictions $g \in \mathcal{H}_C$ then gives

$$\mathbb{P}_\sigma\left(\max_{g\in\mathcal{H}_C}|\theta_g| > \rho\right) \le 2|\mathcal{H}_C|\exp\left(-\frac{m\rho^2}{2}\right). \tag{9}$$

If $X \ge 0$ and $\mathbb{P}(X > t) \le 2b\exp(-t^2/a^2)$ for all $t \ge 0$, where $b \ge e$, then subgaussian concentration gives

$$\mathbb{E}X \le a(2 + \sqrt{\log b}).$$

Apply this with

$$X = \max_{g\in\mathcal{H}_C}|\theta_g|, \qquad a = \sqrt{\frac{2}{m}}, \qquad b = \max\{e, |\mathcal{H}_C|\}.$$

Equation (9) then implies the convenient uniform bound

$$\mathbb{E}_\sigma \max_{g\in\mathcal{H}_C}|\theta_g| \le \frac{8 + 2\sqrt{\log|\mathcal{H}_C|}}{\sqrt{2m}} \le \frac{8 + 2\sqrt{\log(\tau_\mathcal{H}(2m))}}{\sqrt{2m}}.$$

Substitution into (8) proves (6).

It remains to pass convert the expectation to a probability tail bound. Since $G(S)$ is nonnegative, Markov's inequality and (6) give

$$\mathbb{P}_S\left\{G(S) > \frac{8 + 2\sqrt{\log(\tau_\mathcal{H}(2m))}}{\delta\sqrt{2m}}\right\} \le \delta.$$

Therefore the displayed bound holds simultaneously for every $h \in \mathcal{H}$ with probability at least $1 - \delta$. □

If $\mathrm{VCdim}(\mathcal{H}) = d < \infty$, Sauer–Shelah–Perles gives $\tau_\mathcal{H}(2m) \le (2em/d)^d$ once $2m > d + 1$. The right-hand side above then tends to zero for every fixed $\delta$. Hence every finite-VC class has the uniform convergence property, from which learnability follows.

**Theorem 8** (Fundamental Theorem of Statistical Learning)**.** *Let $\mathcal{H}$ be a class of functions from $\mathcal{X}$ to $\{0, 1\}$, with the 0–1 loss. The following statements are equivalent:*

1. *$\mathcal{H}$ has the uniform convergence property;*
2. *every ERM rule is a successful agnostic PAC learner for $\mathcal{H}$;*
3. *$\mathcal{H}$ is agnostic PAC learnable;*
4. *$\mathcal{H}$ is PAC learnable;*
5. *every ERM rule is a successful PAC learner for $\mathcal{H}$;*
6. *$\mathrm{VCdim}(\mathcal{H}) < \infty$.*

*Proof.* 1 implies 2 and 5 via the ERM transfer lemma. 2 and 5 automatically imply 3 and 4 respectively (2 and 5 provides learners). 3 and 4 imply 6 via the No-Free Lunch Theorem (in particular its contrapositive) 6 implies 1 via the theorem 7 from the notes. This gives a directed path between any two statements so their equivalence follows. □

We now provide quantitative versions of statements 1,3, and 4.

**Theorem 9** (Quantitative Fundamental Theorem)**.** *Suppose* $\mathrm{VCdim}(\mathcal{H}) = d < \infty$. *There are absolute constants* $C_1, C_2 > 0$ *such that*

$$C_1 \frac{d + \log(1/\delta)}{\epsilon^2} \leq m_{\mathcal{H}}^{\mathrm{UC}}(\epsilon, \delta) \leq C_2 \frac{d + \log(1/\delta)}{\epsilon^2},$$
$$C_1 \frac{d + \log(1/\delta)}{\epsilon^2} \leq m_{\mathcal{H}}^{\mathrm{agn}}(\epsilon, \delta) \leq C_2 \frac{d + \log(1/\delta)}{\epsilon^2},$$
$$C_1 \frac{d + \log(1/\delta)}{\epsilon} \leq m_{\mathcal{H}}^{\mathrm{PAC}}(\epsilon, \delta) \leq C_2 \frac{d \log(1/\epsilon) + \log(1/\delta)}{\epsilon}.$$

Theorem 7 gives upper bounds which have a polynomial dependence on $\delta$. To obtain the sharp agnostic upper bound one needs to apply a refined a more refined multiscale analysis (i.e. "chaining") of Rademacher complexity of $\mathcal{H}$. In the realizable case, a direct $\epsilon$-net argument gives the displayed upper bound and a weaker dependence $\epsilon$. This last part is not so surprising since the uniform convergence property was only used in proof of agnostic learnability of finite classes.

The definition of $\epsilon$-net provided here is quite different from the one that typically appears in an analysis or optimization text. One should think of it as a sort of "detector" and less so as something that approximates a set.

**Definition 11** ($\epsilon$-net)**.** Let $\mathcal{X}$ be a domain. A set $S \subset \mathcal{X}$ is an $\epsilon$-net for $\mathcal{H} \subset 2^{\mathcal{X}}$ with respect to a distribution $\mathcal{D}$ over $\mathcal{X}$ if

$$\forall h \in \mathcal{H}, \qquad \mathcal{D}(h) \geq \epsilon \quad \Longrightarrow \quad h \cap S \neq \varnothing.$$

Indeed, the definiton of an $\epsilon$-net implies that whenever $h$ classifies too many points as 1, one of those points must be in the net.

**Theorem 10** ($\epsilon$-net theorem)**.** *Let* $\mathcal{A} \subseteq 2^{\mathcal{X}}$ *have VC dimension* $d$. *Fix* $\epsilon \in (0,1)$ *and* $\delta \in (0, 1/4)$. *If*

$$m \geq \frac{8}{\epsilon}\left(2d \log \frac{16e}{\epsilon} + \log \frac{2}{\delta}\right),$$

*then an i.i.d. sample* $S \sim D_X^m$ *is an* $\epsilon$*-net for* $\mathcal{A}$ *with probability at least* $1 - \delta$.

*Proof.* Let

$$B = \{S \subset \mathcal{X} : |S| = m,\ \exists h \in \mathcal{H}, \mathcal{D}(h) \geq \epsilon,\ h \cap S = \varnothing\}$$

be the set of sets which are not $\epsilon$-nets. We need to bound $\mathbb{P}[S \in B]$. Define

$$B' = \{(S,T) \subset \mathcal{X} : |S| = |T| = m,\ \exists h \in \mathcal{H}, \mathcal{D}(h) \geq \epsilon,\ h \cap S = \varnothing,\ |T \cap h| > \epsilon m/2\}.$$

*Claim 1*

$$\mathbb{P}[S \in B] \leq 2\mathbb{P}[(S,T) \in B'].$$

*Proof of Claim 1:* Since $S$ and $T$ are chosen independently we can write

$$\begin{aligned}\mathbb{P}[(S,T) \in B'] &= \mathop{\mathbb{E}}_{(S,T)\sim\mathcal{D}^{2m}}\left[\mathbf{1}_{[(S,T)\in B']}\right] \\ &= \mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}\left[\mathop{\mathbb{E}}_{T\sim\mathcal{D}^m}\left[\mathbf{1}_{[(S,T)\in B']}\right]\right].\end{aligned}$$

Note that $(S,T) \in B'$ implies $S \in B$ and therefore $\mathbf{1}_{[(S,T)\in B']} = \mathbf{1}_{[(S,T)\in B']}\mathbf{1}_{[S\in B]}$, which gives

$$\begin{aligned}\mathbb{P}[(S,T) \in B'] &= \mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}\mathop{\mathbb{E}}_{T\sim\mathcal{D}^m}\mathbf{1}_{[(S,T)\in B']}\mathbf{1}_{[S\in B]} \\ &= \mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}\mathbf{1}_{[S\in B]}\mathop{\mathbb{E}}_{T\sim\mathcal{D}^m}\mathbf{1}_{[(S,T)\in B']}.\end{aligned}$$

Fix some $S$. Then, either $\mathbf{1}_{[S\in B]}=0$ or $S\in B$ and then $\exists h_S$ such that $\mathcal{D}(h_S)\geq\epsilon$ and $|h_S\cap S|=0$. It follows that a sufficient condition for $(S,T)\in B'$ is that $|T\cap h_S|>\epsilon m/2$. Therefore, whenever $S\in B$ we have

$$\mathop{\mathbb{E}}_{T\sim\mathcal{D}^m}\mathbf{1}_{[(S,T)\in B']}\geq\mathop{\mathbb{P}}_{T\sim\mathcal{D}^m}[|T\cap h_S|>\epsilon m/2].$$

But, since we now assume $S\in B$ we know that $\mathcal{D}(h_S)=\rho\geq\epsilon$. Therefore, $|T\cap h_S|$ is a binomial random variable with parameters $\rho$ (probability of success for a single try) and $m$ (number of tries). Chernoff's inequality implies

$$\mathbb{P}[|T\cap h_S|\leq\rho m/2]\leq e^{-\frac{2}{m\rho}(m\rho-m\rho/2)^2}=e^{-m\rho/2}\leq e^{-m\epsilon/2}\leq e^{-d\log(1/\delta)/2}=\delta^{d/2}\leq 1/2.$$

Thus,

$$\mathbb{P}[|T\cap h_S|>\epsilon m/2]=1-\mathbb{P}[|T\cap h_S|\leq\epsilon m/2]\geq 1-\mathbb{P}[|T\cap h_S|\leq\rho m/2]\geq 1/2.$$

Combining all the preceding we *conclude the proof of Claim 1.*

*Claim 2 (Symmetrization):*

$$\mathbb{P}[(S,T)\in B']\leq e^{-\epsilon m/4}\tau_{\mathcal{H}}(2m).$$

*Proof of Claim 2:* To simplify notation, let $\alpha=m\epsilon/2$ and for a sequence $A=(x_1,\ldots,x_{2m})$ let $A_0=(x_1,\ldots,x_m)$. Using the definition of $B'$ we get that

$$\begin{aligned}\mathbb{P}[A\in B'] &= \mathop{\mathbb{E}}_{A\sim\mathcal{D}^{2m}}\max_{h\in\mathcal{H}}\mathbf{1}_{[\mathcal{D}(h)\geq\epsilon]}\mathbf{1}_{[|h\cap A_0|=0]}\mathbf{1}_{[|h\cap A|\geq\alpha]}\\ &\leq \mathop{\mathbb{E}}_{A\sim\mathcal{D}^{2m}}\max_{h\in\mathcal{H}}\mathbf{1}_{[|h\cap A_0|=0]}\mathbf{1}_{[|h\cap A|\geq\alpha]}.\end{aligned}$$

Now, let us define by $\mathcal{H}_A$ the effective number of different hypotheses on $A$, namely, $\mathcal{H}_A=\{h\cap A: h\in\mathcal{H}\}$. It follows that

$$\begin{aligned}\mathbb{P}[A\in B'] &\leq \mathop{\mathbb{E}}_{A\sim\mathcal{D}^{2m}}\max_{h\in\mathcal{H}_A}\mathbf{1}_{[|h\cap A_0|=0]}\mathbf{1}_{[|h\cap A|\geq\alpha]}\\ &\leq \mathop{\mathbb{E}}_{A\sim\mathcal{D}^{2m}}\sum_{h\in\mathcal{H}_A}\mathbf{1}_{[|h\cap A_0|=0]}\mathbf{1}_{[|h\cap A|\geq\alpha]}.\end{aligned}$$

Let $J=\{\mathbf{j}\subset[2m]:|\mathbf{j}|=m\}$. For any $\mathbf{j}\in J$ and $A=(x_1,\ldots,x_{2m})$ define $A_{\mathbf{j}}=(x_{j_1},\ldots,x_{j_m})$. Since the elements of $A$ are chosen i.i.d., we have that for any $\mathbf{j}\in J$ and any function $f(A,A_0)$ it holds that $\mathbb{E}_{A\sim\mathcal{D}^{2m}}[f(A,A_0)]=\mathbb{E}_{A\sim\mathcal{D}^{2m}}[f(A,A_{\mathbf{j}})]$. Since this holds for any $\mathbf{j}$ it also holds for the expectation of $\mathbf{j}$ chosen at random from $J$. In particular, it holds for the function $f(A,A_0)=\sum_{h\in\mathcal{H}_A}\mathbf{1}_{[|h\cap A_0|=0]}\mathbf{1}_{[|h\cap A|\geq\alpha]}$. We therefore obtain that

$$\begin{aligned}\mathbb{P}[A\in B'] &\leq \mathop{\mathbb{E}}_{A\sim\mathcal{D}^{2m}}\mathop{\mathbb{E}}_{\mathbf{j}\sim J}\sum_{h\in\mathcal{H}_A}\mathbf{1}_{[|h\cap A_{\mathbf{j}}|=0]}\mathbf{1}_{[|h\cap A|\geq\alpha]}\\ &= \mathop{\mathbb{E}}_{A\sim\mathcal{D}^{2m}}\sum_{h\in\mathcal{H}_A}\mathbf{1}_{[|h\cap A|\geq\alpha]}\mathop{\mathbb{E}}_{\mathbf{j}\sim J}\mathbf{1}_{[|h\cap A_{\mathbf{j}}|=0]}.\end{aligned}$$

Now, fix some $A$ s.t. $|h\cap A|\geq\alpha$. Then, $\mathbb{E}_{\mathbf{j}}\mathbf{1}_{[|h\cap A_{\mathbf{j}}|=0]}$ is the probability that when choosing $m$ balls from a bag with at least $\alpha$ red balls, we will never choose a red ball. This probability is at most

$$(1-\alpha/(2m))^m=(1-\epsilon/4)^m\leq e^{-\epsilon m/4}.$$

We therefore get that

$$\mathbb{P}[A \in B'] \leq \mathop{\mathbb{E}}_{A\sim\mathcal{D}^{2m}} \sum_{h\in\mathcal{H}_A} e^{-\epsilon m/4} \leq e^{-\epsilon m/4} \mathop{\mathbb{E}}_{A\sim\mathcal{D}^{2m}} |\mathcal{H}_A|.$$

Using the definition of the growth function we *conclude the proof of Claim 2.*

*Completing the Proof:* By Sauer's lemma we know that $\tau_{\mathcal{H}}(2m) \leq (2em/d)^d$. Combining this with the two claims we obtain that

$$\mathbb{P}[S \in B] \leq 2(2em/d)^d e^{-\epsilon m/4}.$$

We would like the right-hand side of the inequality to be at most $\delta$; that is,

$$2(2em/d)^d e^{-\epsilon m/4} \leq \delta.$$

Rearranging, we obtain the requirement

$$m \geq \frac{4}{\epsilon}(d\log(2em/d) + \log(2/\delta)) = \frac{4d}{\epsilon}\log(m) + \frac{4}{\epsilon}(d\log(2e/d) + \log(2/\delta)).$$

Using Lemma A.2, a sufficient condition for the preceding to hold is that

$$m \geq \frac{16d}{\epsilon}\log\left(\frac{8d}{\epsilon}\right) + \frac{8}{\epsilon}(d\log(2e/d) + \log(2/\delta)).$$

A sufficient condition for this is that

$$\begin{aligned} m &\geq \frac{16d}{\epsilon}\log\left(\frac{8d}{\epsilon}\right) + \frac{16}{\epsilon}\left(d\log(2e/d) + \tfrac{1}{2}\log(2/\delta)\right) \\ &= \frac{16d}{\epsilon}\left(\log\left(\frac{8d\,2e}{d\epsilon}\right)\right) + \frac{8}{\epsilon}\log(2/\delta) \\ &= \frac{8}{\epsilon}\left(2d\log\left(\frac{16e}{\epsilon}\right) + \log\left(\frac{2}{\delta}\right)\right). \end{aligned}$$

and this concludes our proof. □

We now show that the ERM rule will always output the realizable target whenever $S$ is an $\epsilon$-net (thus establishing the upper bound on $m_{\mathcal{H}}^{\text{PAC}}$). We first consider the case where the realizable target $h^*$ is exactly $\emptyset$ (i.e. it classifies everything as 0). Because $S$ is an $\epsilon$-net any $h$ satisfying $\mathcal{D}(h) \geq \epsilon$ will misclassify some point in $S$. Therefore the ERM rule won't return any such function. Therefore the returned function will satisfy $L_{\mathcal{D}}(h) = \mathcal{D}(h) < \epsilon$.

Next, we consider the case where $h^*$ is arbitrary. Consider the hypothesis class

$$\mathcal{H}' := \{h \otimes h^* : h \in \mathcal{H}\}.$$

Here $\otimes$ denotes the XOR operator ($1 \otimes 0 = 0 \otimes 1 = 1, 1 \otimes 1 = 0 \otimes 0 = 0$) and on a pair of binary strings the operation is applied entrywise. the maps $h \mapsto h \otimes h^*$ gives a bijection between $\mathcal{H}$ and $\mathcal{H}'$ that maps $h^*$ to $\emptyset$. Most importantly is that

$$\mathbf{1}(h(x) = h^*(x)) = \mathbf{1}((h \otimes h^*)(x) = \emptyset(x)) \qquad L_{\mathcal{D},h^*}(h) = L_{\mathcal{D},\emptyset}(h \otimes h^*) \text{ for all } h \in \mathcal{H}, x \in \mathcal{X}.$$

In particular the ERM rule applied to $H$ cannot return a function $h$ for which $L_{\mathcal{D},h^*}(h) \geq \epsilon$ as this would imply that $\epsilon \leq L_{\mathcal{D},\emptyset}(h \otimes h^*) = \mathcal{D}(h \otimes h^*)$ meaning $S$ contained a sample $(x, h^*(x))$ satisfying $\mathbf{1}(h(x) = h^*(x)) = \mathbf{1}((h \otimes h^*)(x) = \emptyset(x)) = 0$, contradicting the ERM rule. Therefore the returned predictor must have true risk below $\epsilon$. This proves the realizable upper bound in the quantitative theorem.

For proofs of the lower bounds we refer the reader to Chapter 28 of the UML textbook. To prove the sharp agnostic upper bound, we next bound the Rademacher complexity of a binary hypothesis class in terms of its VC dimension and apply this bound to empirical risk minimization.

# 7 Rademacher complexity and generalization

VC dimension measures how many distinct labeling a hypothesis class can have on a subset of the domain. Rademacher complexity instead measures how well a class can correlate with independent random signs (e.g. "noise"). The motivation comes from the fact that a function $f \in \mathcal{F}$ and iid random variables $X, X_1, \cdots, X_m, X_1', \cdots, X_m'$ and iid rademacher variables $\sigma_1, \cdots, \sigma_m$ satisfy

$$\begin{aligned}
\sup_{f\in\mathcal{F}}\left|\mathbb{E}f(X) - \frac{1}{m}\sum_{i=1}^m f(X_i)\right| &= \sup_{f\in\mathcal{F}}\left|\frac{1}{m}\sum_{i=1}^m \mathbb{E}f(X_i') - f(X_i)\right| \\
&\leq \mathbb{E}\sup_{f\in\mathcal{F}}\left|\frac{1}{m}\sum_{i=1}^m f(X_i') - f(X_i)\right| \\
&= \mathbb{E}\sup_{f\in\mathcal{F}}\left|\frac{1}{m}\sum_{i=1}^m \sigma_i(f(X_i') - f(X_i))\right| \\
&= \mathbb{E}\sup_{f\in\mathcal{F}}\left|\frac{1}{m}\sum_{i:\epsilon_i=1} f(X_i'') - \sum_{i:\epsilon_i=-1} f(X_i'')\right| \\
&\leq 2\mathbb{E}\sup_{f\in\mathcal{F}}\left|\frac{1}{m}\sum_{i=1}^m \sigma_i f(X_i)\right|
\end{aligned}$$

In Statistical Learning Theory we are interesting in controlling $|L_{\mathcal{D}}(f) - L_S(f)|$ (the difference between true and empirical risk of $f$). Because a learning algorithm computes $f$ from $S$ no randomness from $S$ can be used to show concentration of $|L_{\mathcal{D}}(f) - L_S(f)|$ if we first condition on it. Instead one can treat $S$ as random and consider the worst case approximation over all potentially learnable hypothesis. The Rademacher complexity can therefore be viewed as a measurement of the uniform convergence property of the hypothesis. If the class is too expressive then the Rademacher complexity will be large, meaning the uniform convergence property need not hold. If it is small then the class will indeed have the uniform convergence property (which we'll see later). As a comment, having Rademacher complexity tending to 0 with the sample size actually implies that the VC-dimension (assuming classification) is finite (we will show this later). This is perhaps not so surprising since VC dimension determines learnability. On the other hand, the Rademacher complexity can be relatively large when $S$ is smaller than the VC dimension. In addition, Rademacher complexity is well defined whenever $\sup_{f\in\mathcal{F}}|f(X)|$ is integrable and thus can be used to study learnability of hypothesis classes with non 0-1 loss. One important alternative is Lipschitz-loss. In particular the loss of a random sample does not need to be a bounded random variable. Applying the arguments from finite-class and finite-VC dimension to this setting however will require more general purpose concentration inequalities, which we'll introduce in subsequent sections.

Our goal in this section is to prove a weaker version of the agnostic upper bound in Theorem 9:

$$m_{\mathcal{H}}^{\text{agn}}(\epsilon,\delta) \leq C\frac{d\log(1/\delta)}{\epsilon^2}.$$

The essential estimate will be $R(\mathcal{H}\circ S_X) \leq C\sqrt{d/m}$, where $S_X = (x_1,\ldots,x_m)$ and $d = \text{VCdim}(\mathcal{H})$. The tight bound (the numerator being $d+\log(1/\delta)$ )will require a method knowing as "chaining" which we'll introduce some time later. Throughout this section, function classes are nonempty and bounded ($\sup_{f\in\mathcal{F}}\|f\|_\infty) < \infty$), samples are i.i.d., and all suprema and learning rules under consideration are assumed measurable. For example, pointwise measurable classes permit the suprema below to

be taken over countable subclasses. We write $Pf = \mathbb{E}_{Z\sim D} f(Z)$ and $P_m f = m^{-1}\sum_{i=1}^m f(Z_i)$; the symmetrization statements are used for uniformly bounded classes.

For a sample $S = (z_1, \ldots, z_m)$ and a class $\mathcal{F}$ of real-valued functions on $\mathcal{Z}$, write

$$\mathcal{F} \circ S = \{(f(z_1), \ldots, f(z_m)) : f \in \mathcal{F}\} \subseteq \mathbb{R}^m.$$

For $A \subseteq \mathbb{R}^m$ we write

$$R(A) = \frac{1}{m}\mathbb{E}_\sigma \sup_{a\in A} \sum_{i=1}^m \sigma_i a_i, \qquad R^+(A) = \frac{1}{m}\mathbb{E}_\sigma \sup_{a\in A} \left|\sum_{i=1}^m \sigma_i a_i\right|$$

where $\sigma_1, \ldots, \sigma_m$ are independent random variables with $\mathbb{P}(\sigma_i = 1) = \mathbb{P}(\sigma_i = -1) = 1/2$. The empirical Rademacher complexity of $\mathcal{F}$ on $S$ is $R(\mathcal{F}\circ S)$, and its expected complexity is $\mathbb{E}_S R(\mathcal{F}\circ S)$.

**Lemma 5** (Symmetrization)**.** *Let $A \subset \mathbb{R}^m$. Then*

$$R(A) \le R^+(A) \le 2R(A).$$

*Furthermore let*

$$\mathrm{Rep}_D(\mathcal{F}, S) = \sup_{f\in\mathcal{F}}(Pf - P_m f), \qquad \mathrm{Rep}_D^+(\mathcal{F}, S) = \sup_{f\in\mathcal{F}}|Pf - P_m f|.$$

*Then*

$$\mathbb{E}_S \mathrm{Rep}_D(\mathcal{F}, S) \le 2\mathbb{E}_S R(\mathcal{F}\circ S) \qquad \mathbb{E}_S \mathrm{Rep}_D^+(\mathcal{F}, S) \le 4\mathbb{E}_S R(\mathcal{F}\circ S).$$

*Proof.* For the first claim we have that

$$\begin{aligned} R(A) \le R^+(A) &= \frac{1}{m}\mathbb{E}_\sigma \sup_{a\in A}\left|\sum_{i=1}^m \sigma_i a_i\right| \\ &= \frac{1}{m}\mathbb{E}_\sigma \sup_{a\in A\cup(-A)} \sum_{i=1}^m \sigma_i a_i = R(A\cup(-A)) \\ &\le R(A) + R(-A) = 2R(A). \end{aligned}$$

The second equality uses the fact that either $\sum_{i=1}^m \sigma_i a_i$ or $\sum_{i=1}^m \sigma_i(-a_i)$ is non-negative, the first inequality uses the fact that $R$ is subadditive and the last equality uses the fact that $\sum_{i=1}^m \sigma_i a_i$ and $\sum_{i=1}^m \sigma_i(-a_i)$ have the same distribution.

For the second claim we first note that

$$\mathrm{Rep}_D(\mathcal{F}, S) = \mathrm{Rep}_D(\mathcal{F}', S), \qquad \mathrm{Rep}_D^+(\mathcal{F}, S) = \mathrm{Rep}_D^+(\mathcal{F}', S),$$

where $\mathcal{F}' = \{f - Pf \ : \ f \in \mathcal{F}\}$. This is because the map $f \mapsto Pf - P_m f$ is linear in $f$ and sends constants to 0. Thus we have

$$\begin{aligned} \mathbb{E}_S \mathrm{Rep}_D(\mathcal{F}, S) &\le \mathbb{E}_S \mathrm{Rep}_D^+(\mathcal{F}, S) = \mathbb{E}_S \mathrm{Rep}_D^+(\mathcal{F}', S) \\ &= \mathbb{E}_S \sup_{f\in\mathcal{F}'}\left|\frac{1}{m}\sum_{i=1}^m f(z_i)\right| \\ &= \mathbb{E}_S \sup_{f\in\mathcal{F}'}\left|\frac{1}{m}\sum_{i=1}^m (f(z_i) - \mathbb{E}_{z_i'} f(z_i'))\right| \\ &\le \mathbb{E}_S\mathbb{E}_{S'} \sup_{f\in\mathcal{F}'}\left|\frac{1}{m}\sum_{i=1}^m (f(z_i) - f(z_i'))\right| \\ &= \mathbb{E}_S\mathbb{E}_{S'}\mathbb{E}_\sigma \sup_{f\in\mathcal{F}'}\left|\frac{1}{m}\sum_{i=1}^m \sigma_i(f(z_i) - f(z_i'))\right|. \end{aligned}$$

Conditional on $S, S'$ we define $A_{S,S'} = \{(f(z_1) - f(z_1'), \cdots, f(z_n) - f(z_n')) : f \in \mathcal{F}'\}$. Clearly $R(A_{S,S'}) \leq R(\mathcal{F}' \circ S) + R((-\mathcal{F}') \circ S')$. Therefore

$$\begin{aligned}
&\mathbb{E}_S \mathbb{E}'_S \mathbb{E}_\sigma \sup_{f \in \mathcal{F}'} \left| \frac{1}{m} \sum_{i=1}^m \sigma_i (f(z_i) - f(z_i')) \right| \\
&\quad = \mathbb{E}_S \mathbb{E}_{S'} R^+(A_{S,S'}) \leq \mathbb{E}_S \mathbb{E}_{S'} 2R(A_{S,S'}) \\
&\quad \leq \mathbb{E}_S \mathbb{E}_{S'} 2[R(\mathcal{F}' \circ S) + R(-\mathcal{F}' \circ S')] \\
&\quad = 4\mathbb{E}_S R(\mathcal{F}' \circ S) = 4\mathbb{E}_S R(\mathcal{F} \circ S).
\end{aligned}$$

This establishes $\mathbb{E}_S \mathrm{Rep}_D^+(\mathcal{F}, S) \leq 4\mathbb{E}_S R(\mathcal{F} \circ S)$. For the other inequality the same argument gives $\mathbb{E}_S \mathrm{Rep}_D(\mathcal{F}, S) \leq \mathbb{E}_S \mathbb{E}_{S'} R(A_{S,S'}) \leq 2\mathbb{E}_S R(\mathcal{F} \circ S)$. □

The following lemma show that $R(\mathcal{F} \circ S)$ scales like $\frac{\mathrm{rad}(\mathcal{F} \circ S)\sqrt{\log |\mathcal{F}|}}{|S|}$, where $\mathrm{rad}(\mathcal{F} \circ S)$ denote the radius of $\mathcal{F} \circ S$ when viewed as a subset of $\mathbb{R}^m$.

**Lemma 6** (Massart's lemma)**.** *Let* $A = \{a_1, \ldots, a_N\} \subset \mathbb{R}^m$ *and* $\bar{a} = N^{-1} \sum_{j=1}^N a_j$. *Define* $\mathrm{rad}(A) = \max_{a \in A} \|a - \bar{a}\|_2$. *Then*

$$R(A) \leq \mathrm{rad}(A) \frac{\sqrt{2 \log N}}{m} + \frac{\|\bar{a}\|_2}{m}.$$

*Proof.* For convenience we write $r := \mathrm{rad}(A)$. We first consider the case where $\bar{a} = 0$. If $r = 0$ then $A$ is a singleton and, since $\bar{a} = 0$ is simply $\{0\}$. Clearly then $R(A) = 0$ and we're done. Thus we assume that $r > 0$. For a fixed vector $v$, independence and the inequality $\cosh u \leq e^{u^2/2}$ give

$$\mathbb{E}_\sigma e^{\lambda \sum_i \sigma_i v_i} = \prod_i \cosh(\lambda v_i) \leq e^{\lambda^2 \|v\|_2^2 / 2}.$$

Indeed, $(\log \cosh u)' = \tanh u \leq u$ for $u \geq 0$, and both functions are even. Therefore

$$R(A) \leq \frac{\log N}{\lambda} + \frac{\lambda r^2}{2}.$$

Taking $\lambda = \sqrt{2 \log N}/r$ then gives

$$\begin{aligned}
mR(A) &= \mathbb{E}_\sigma \max_{a \in A} \sum_{i=1}^m \sigma_i a_i = \mathbb{E}_\sigma \frac{1}{\lambda} \left( \max_{a \in A} \log \exp \sum_{i=1}^m \lambda \sigma_i a_i \right) \\
&\leq \mathbb{E}_\sigma \frac{1}{\lambda} \left( \log \max_{a \in A} \exp \sum_{i=1}^m \lambda \sigma_i a_i \right) \leq \frac{1}{\lambda} \log \mathbb{E}_\sigma \left( \max_{a \in A} \exp \sum_{i=1}^m \lambda \sigma_i a_i \right) \\
&= \frac{1}{\lambda} \log \mathbb{E}_\sigma \left( \sum_{a \in A} \exp \sum_{i=1}^m \lambda \sigma_i a_i \right) \leq \frac{1}{\lambda} \log \left( N \max_{a \in A} \mathbb{E}_\sigma \left( \exp \sum_{i=1}^m \lambda \sigma_i a_i \right) \right) \\
&\frac{1}{\lambda} \log \left( N \exp \left( \lambda^2 r^2 / 2 \right) \right) = r\sqrt{2 \log N}
\end{aligned}$$

Thus $R(A) \leq \frac{r\sqrt{2 \log N}}{m}$. In the case where $\bar{a} \neq 0$ we note that $R(A) = R(A - \bar{a})$ since

$$\mathbb{E}_\sigma \sup_{a \in A} \sigma \cdot a = \mathbb{E}_\sigma \sup_{a \in A} \sigma \cdot ((a - \bar{a}) + \bar{a}) = \mathbb{E}_\sigma (\sigma \cdot \bar{a} + \sup_{a \in A} \sigma \cdot (a - \bar{a})) = \mathbb{E}_\sigma \sup_{a \in A} \sigma \cdot (a - \bar{a})$$

The result then follows from the centered case. □

As a comment, if one instead uses the inequality $R(A) \leq R(A - \bar{a}) + R(\bar{a})$ one can bound $R(\bar{a})$ by $\|\bar{a}\|_2/m$ using only the facts that the $\sigma_i$ are pairwise independent, standardized and monotonicty of $\ell_p$ norms. Although that gives a weaker bound, that bound is robust to the following scenario: Suppose that $A$ is replaced with $A - b$ where $A$ is still deterministic but $b$ is a function of $\sigma$ that satisfies $(\sigma \cdot (\bar{a} - b)) = |(\sigma \cdot \bar{a})|$. One can think of this situation as forcing a weak alignment between $A$ and $\sigma$. The corresponding Rademacher complexity $R'$ would satisfy $R'(A) \leq R(A) + \frac{\|\bar{a}\|_2}{m}$. This is also related to changing the definition of $R(A)$ so as to require that $(\sigma \cdot \bar{a}) \geq 0$ (hence $\sigma$ is only uniform over the choice of Rademacher vectors which satisfy this condition). In that case one can show that the corresponding Rademacher complexity $R''$ satisfies $R''(A) \leq R(A) + \frac{r\sqrt{2}}{m} + \frac{\|\bar{a}\|_2}{m}$, giving a bound which is in fact worse that the adversarial shift. Lastly, there are examples of $A$ with $\bar{a} \neq 0$, but made arbitrarily small, for which $R''(A) \geq R(A) + \frac{r}{m}$. In particular $R''(A) - R(A)$ can be arbitrarily large compared to $R'(A) - R(A)$.

**Lemma 7** (Contraction)**.** *For each $i$, let $\phi_i : \mathbb{R} \to \mathbb{R}$ be $\rho$-Lipschitz. For $a \in \mathbb{R}^m$, put $\phi(a) = (\phi_1(a_1), \ldots, \phi_m(a_m))$. Then*

$$R(\phi \circ A) \leq \rho R(A).$$

*Proof.* Without loss of generality we may assume that $\rho = 1$ since $\rho^{-1}\phi$ is 1-Lipschitz and $R(\rho^{-1}\phi \circ A) = \rho^{-1}R(\phi \circ A)$. We have

$$\begin{aligned}
R(\phi \circ A) = \mathbb{E}_\sigma \sup_{a \in A} \sum_{i=1}^m \sigma_i \phi_i(a) &= \frac{1}{2}\left(\mathbb{E}_\sigma \sup_{a \in A} \phi_1(a) + \sum_{i=2}^m \sigma_i \phi_i(a) + \sup_{b \in A} -\phi_1(b) + \sum_{i=2}^m \sigma_i \phi_i(b)\right) \\
&= \frac{1}{2}\left(\mathbb{E}_\sigma \sup_{a,b \in A} (\phi_1(a) - \phi_1(b)) + \sum_{i=2}^m \sigma_i(\phi_i(a) + \phi_i(b))\right) \\
&\leq \frac{1}{2}\left(\mathbb{E}_\sigma \sup_{a,b \in A} |a_1 - b_1| + \sum_{i=2}^m \sigma_i(\phi_i(a) + \phi_i(b))\right) \\
&= \frac{1}{2}\left(\mathbb{E}_\sigma \sup_{a,b \in A} (a_1 - b_1) + \sum_{i=2}^m \sigma_i(\phi_i(a) + \phi_i(b))\right) \\
&\leq \frac{1}{2}\mathbb{E}_\sigma \sup_{a \in A} a_1 + \sum_{i=2}^m \sigma_i(\phi_i(a)) + \frac{1}{2}\mathbb{E}_\sigma \sup_{b \in A} -b_1 + \sum_{i=2}^m \sigma_i(\phi_i(b)) \\
&= \mathbb{E}_\sigma \sup_{a \in A} \sigma_1 a_1 + \sum_{i=2}^m \sigma_i(\phi_i(a)).
\end{aligned}$$

Iterating the argument the argument on the remaining coordinates gives $R(\phi \circ A) \leq R(A)$. □

We will need the following lemma to show that concentration of $|L_D(f) - L_S(f)|$ reduces to concentration of $R(\mathcal{F} \circ S)$.

**Lemma 8** (McDiarmid's inequality)**.** *Let $V$ be a set and let $f : V^m \to \mathbb{R}$ be a function such that, for some $c > 0$, every $i \in \{1, \ldots, m\}$, and all $x_1, \ldots, x_m, x_i' \in V$,*

$$\left|f(x_1, \ldots, x_m) - f(x_1, \ldots, x_{i-1}, x_i', x_{i+1}, \ldots, x_m)\right| \leq c.$$

*Let $X_1, \ldots, X_m$ be independent random variables taking values in $V$. For every $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$|f(X_1, \ldots, X_m) - \mathbb{E}f(X_1, \ldots, X_m)| \leq c\sqrt{\frac{m}{2}\log\frac{2}{\delta}}.$$

This is a standard result in concentration inequalities. You would likely see this in a course on Probabilistic Combinatorics. One proof involves realizing $f(X_1, \cdots, X_m)$ as the initialization of a doob-martingale and applying the laplace method to control it's large deviation behavior along the process.

Combining this inequality with symmetrization gives both an expected-complexity bound and a bound that can be evaluated on the observed sample.

**Theorem 11** (Data-dependent Rademacher bound)**.** *Suppose $|\ell(h, z)| \le c$ for all $h \in \mathcal{H}$ and $z \in \mathcal{Z}$. For every $\delta \in (0, 1)$, with probability at least $1 - \delta$, simultaneously for all $h \in \mathcal{H}$,*

$$L_D(h) - L_S(h) \le 2\mathbb{E}_{S'} R(\ell \circ \mathcal{H} \circ S') + c\sqrt{\frac{2\log(2/\delta)}{m}}.$$

*Also, with probability at least $1 - \delta$, simultaneously for all $h$,*

$$L_D(h) - L_S(h) \le 2R(\ell \circ \mathcal{H} \circ S) + 4c\sqrt{\frac{2\log(4/\delta)}{m}}.$$

*Moreover, for any fixed comparator $h^* \in \mathcal{H}$, with probability at least $1 - \delta$,*

$$L_D(\widehat{h}_S) - L_D(h^*) \le 2R(\ell \circ \mathcal{H} \circ S) + 5c\sqrt{\frac{2\log(8/\delta)}{m}}.$$

*Proof.* If $c = 0$, all losses vanish and the conclusions are immediate. Assume $c > 0$. Put $F = \ell \circ \mathcal{H}$ and $G(S) = \sup_{f \in F}(Pf - P_m f)$. Replacing one observation changes $P_m f$ by at most $2c/m$, uniformly in $f$, so it also changes $G$ by at most $2c/m$. McDiarmid's inequality (Lemma 8), with coordinate bound $2c/m$, and symmetrization imply, for $\eta \in (0, 1)$, with probability at least $1 - \eta$,

$$G(S) \le \mathbb{E}G(S) + c\sqrt{\frac{2\log(2/\eta)}{m}} \le 2\mathbb{E}_S R(F \circ S) + c\sqrt{\frac{2\log(2/\eta)}{m}}.$$

This proves the first assertion. The function $S \mapsto R(F \circ S)$ also changes by at most $2c/m$ under one replacement: for each fixed sign vector, every signed sample average changes by at most $2c/m$, and taking the supremum and then the expectation preserves this bound. Applying Lemma 8 to this function gives, with probability at least $1 - \delta/2$,

$$\mathbb{E}_{S'} R(F \circ S') \le R(F \circ S) + c\sqrt{\frac{2\log(4/\delta)}{m}}.$$

Intersect this event with the preceding bound on $G$, taking $\eta = \delta/2$. A union bound gives, with probability at least $1 - \delta$,

$$G(S) \le 2R(F \circ S) + 3c\sqrt{\frac{2\log(4/\delta)}{m}},$$

which implies the stated bound with constant $4c$.

Finally, decompose the excess risk as

$$\begin{aligned} L_D(\widehat{h}_S) - L_D(h^*) = [L_D(\widehat{h}_S) - L_S(\widehat{h}_S)] + [L_S(\widehat{h}_S) - L_S(h^*)] \\ + [L_S(h^*) - L_D(h^*)]. \end{aligned}$$

The middle term is nonpositive by empirical optimality. The second assertion, with failure probability $\delta/2$, bounds the first term by $2R(F \circ S) + 4c\sqrt{2\log(8/\delta)/m}$. Hoeffding's inequality for the fixed function $\ell(h^*, \cdot) \in [-c, c]$ bounds the last term by $c\sqrt{2\log(4/\delta)/m}$, also with failure probability at most $\delta/2$. A union bound and $\log(4/\delta) \le \log(8/\delta)$ give the result. □

For a finite prediction class $\mathcal{G}$ with $|g(x)| \le B$ and a $\rho$-Lipschitz loss, Massart's lemma and contraction give

$$R(\ell \circ \mathcal{G} \circ S) \le 2\rho B \sqrt{\frac{2\log|\mathcal{G}|}{m}}.$$

For infinite classes we instead bound the complexity directly.

We now return to binary classification. Let $\mathcal{H} \subseteq \{0,1\}^{\mathcal{X}}$ have VC dimension $d$. For a fixed labeled sample, set

$$A_X = \{(h(x_i))_{i=1}^m : h \in \mathcal{H}\}, \qquad A_S = \{(\mathbf{1}\{h(x_i) \neq y_i\})_{i=1}^m : h \in \mathcal{H}\}.$$

In fact, the loss class has exactly the same empirical Rademacher complexity as the prediction class. Since $\mathbf{1}\{h(x_i) \neq y_i\} = y_i + (1-2y_i)h(x_i)$, we have

$$\begin{aligned} R(A_S) &= \frac{1}{m}\mathbb{E}_\sigma \sum_i \sigma_i y_i + \frac{1}{m}\mathbb{E}_\sigma \sup_{h\in\mathcal{H}} \sum_i \sigma_i (1-2y_i)h(x_i) \\ &= R(A_X). \end{aligned}$$

Here the first expectation vanishes and the signs $\sigma_i(1-2y_i)$ are again independent Rademacher variables.

For $1 \le d \le m$, Sauer's lemma and Massart's lemma give

$$R(A_S) = R(A_X) \le \sqrt{\frac{2\log|A_X|}{m}} \le \sqrt{\frac{2d\log(em/d)}{m}}.$$

As mentioned earlier, the sharp bound is obtained by showing that $R(A_s) = O(\sqrt{d/m})$. This involves the chaining method, which we will see in a later section.

# 8 Concentration Inequalities: An Introduction

We now transition to the second part of the course, which is focused on concentration inequalities. The overall treatment will be broad but we begin by giving some motivation for studying it. Recall that in the previous sections on Learning Theory our general strategy for showing PAC-learnability and agnostic PAC-learnability (with sample complexity guarantees) of hypothesis classes was centered around proving that a learnable hypothesis class had the uniform convergence property: For every $\epsilon, \delta$ there exists $m(\epsilon,\delta)$ such that whenever $m \ge m(\epsilon,\delta)$ one has

$$\mathbb{P}_{S\sim\mathcal{D}^m}\left(\sup_{f\in\mathcal{F}} |L_{\mathcal{D}}(f) - L_S(f)| > \epsilon\right) < \delta$$

In our previous analyses we viewed $L_S(f)$ as an average of iid random variables and $L_{\mathcal{D}}(F)$ as its mean, allowing us to control the deviation of the quantity for a single $f$. Union bounding over all admissible $f$ gave us the uniform statement. As a comment, note that the uniform statement requires sufficiently strong control of the deviation (otherwise the choice of $\epsilon$ could be trivial). In our analasis of the learnability of binary classification we used sub-gaussian concentration of averages of iid bounded variables and assumed a 0-1 loss (although in the section on Rademacher complexity we saw that Lipschitz losses could potentially be used). It turns out that these assumptions can be generalized (the type of loss and the type of random variable).

In this section we will introduce the theory by proving 3 results:

1. Hoeffding's inequality
2. Azuma-Hoeffding inequality
3. McDiarmid's inequality

All 3 results are largely based on the following technical lemma:

**Lemma 9** (Hoeffding's Lemma)**.** *Let $X$ be an integrable random variable satisfying $a \leq X - \mathbb{E}[X] \leq b$ almost surely. Then*

$$\mathbb{E}\exp(\lambda X - \mathbb{E}[X]) \leq \exp(\lambda^2(b-a)^2/8) \qquad \forall \lambda \in \mathbb{R}.$$

Note that the boundedness assumption is typically of the form $a \leq X \leq b$, but if you shift everything by $-\mathbb{E}[X]$ then the difference between the upper and lower limits is still $(b-a)$. This version of the assumption will make more sense when we see the proof of Azuma-Hoeffding.

*Proof.* If $\lambda = 0$ the inequality is equivalent to $1 \leq 1$. Thus we may assume that $\lambda \neq 0$. We first note that

$$a \leq X - \mathbb{E}[X] \leq b \implies \mathbb{E}[a] \leq \mathbb{E}[X - \mathbb{E}[X]] \leq \mathbb{E}[b] \implies a \leq 0 \leq b.$$

In particular either $a < b$ or $a = b = 0 = X - \mathbb{E}[X]$ (with the last equality holding almost surely). In the latter case the desired inequality again reduces to $1 \leq 1$. Thus we're left with the case that $a < b$.

For this case we recall that the map $x \mapsto \exp(\lambda x)$ is a convex function in $x$, regardless of $\lambda$. Since $x - \mathbb{E}[X] = at + b(1-t) \iff t = \frac{b-(x-\mathbb{E}[X])}{b-a}$), where $t \in [0,1]$, it follows from convexity of the map that

$$\begin{aligned}
\exp\left(\lambda(x - \mathbb{E}[X])\right) &= \exp\left(\lambda a \frac{b-(x-\mathbb{E}[X])}{b-a} + \lambda b \frac{(x-\mathbb{E}[X])-a}{b-a}\right) \\
&\leq \frac{b-(x-\mathbb{E}[X])}{b-a}\exp(\lambda a) + \frac{(x-\mathbb{E}[X])-a}{b-a}\exp(\lambda b) \\
\implies \mathbb{E}\exp(\lambda(X - \mathbb{E}[X])) &\leq \frac{b}{b-a}\exp(\lambda a) + \frac{-a}{b-a}\exp(\lambda b).
\end{aligned}$$

Now we reparametrize the resulting upper bound. We define the following parameters:

$$L = b - a, p := \frac{b}{L}, s = \lambda L.$$

Then

$$\frac{b}{b-a}\exp(\lambda a) + \frac{-a}{b-a}\exp(\lambda b) = p\exp(-s(1-p)) + (1-p)\exp(sp) = \exp(sp)(1 - p + p\exp(-s)).$$

For convenience we define $f(s) := (1 - p + p\exp(-s))$. Now define $h(s) := \log(\exp(sp)f(s)) = sp + \log(f(s))$. Notice that $h(s)$ is exactly the same as the log of the reparametrized upper bound. Therefore we would be finished if we showed that $h(s) \leq s^2/8$ for all $s \in \mathbb{R}$. To that end we compute some things:

- $h(0) = 0 \cdot p + \log(f(0)) = 0 + 1\log(1) = 0.$
- $h'(s) = p + \frac{f'(s)}{f(s)} = p - \frac{p\exp(-s)}{f(s)}$. Therefore $h'(0) = p - \frac{p}{1} = 0.$

- Observe that $f''(s) = -f'(s)$ and $f(s) = 1 - p - f'(s)$. Thus we have

$$h''(s) = \frac{f''(s)f(s) - f'(s)^2}{f(s)^2} = \frac{f''(s)(f(s) + f'(s))}{f(s)^2} = \frac{(-f'(s))(1-p)}{f(s)^2} = \left(1 - \frac{1-p}{f(s)}\right)\left(\frac{1-p}{f(s)}\right)$$

  Now observe that the map $x \mapsto x(1-x)$ is a negative parabola whose maximum is obtained at $x = 1/2$ with value $1/4$. Thefore $h''(s) \leq 1/4$ for all $s \in \mathbb{R}$. Integration then gives

$$h(s) = h(0) + \int_0^s h'(t)\ dt = 0 + \int_0^s \left(h'(0) + \int_0^t h''(v)\ dv\right)\ dt \leq \int_0^s \int_0^t 1/4\ dv\ dt = s^2/8.$$

  Having shown $h(s) \leq s^2/8$ we conclude as desired.

□

As a remark we note that the proof of Hoeffding's Lemma implies that

$$\frac{b}{b-a}\exp(\lambda a) + \frac{-a}{b-a}\exp(\lambda b) \leq \exp(\lambda^2(b-a)^2/8) \qquad \forall a < b. \tag{10}$$

Indeed, one can check that the parametrization and subsequent convexity analysis only required that $a < b$. In the proof of Azuma-Hoeffding we will actually need a conditional version of Hoeffding's Lemma (where $\mathbb{E}[X]$ is replaced with $\mathbb{E}[X|\mathcal{F}]$ for some sub sigma-algebra $\mathcal{F}$ coming from a filtration). The proof will follow almost immediately from (10). For the statement we will assume that all of our random variables exist over a common probability space $(\mathbb{P}, \Omega, \Sigma)$

**Corollary 2** (Conditional Hoeffding's Lemma)**.** *Let $X$ be an integrable random variable and let $\mathcal{F} \subset \Sigma$ be a sub-sigma algebra. Suppose $A \leq X - \mathbb{E}[X|\mathcal{F}] \leq B$ almost surely, where $A, B$ are random variables that are $\mathcal{F}$ measurable. Then*

$$\mathbb{E}[\exp(\lambda X - \mathbb{E}[X|\mathcal{F}])|\mathcal{F}] \leq \exp(\lambda^2(B-A)^2/8) \text{ almost surely} \qquad \forall \lambda \in \mathbb{R}.$$

Note that Hoeffding's Lemma is an immediate consequence of this by taking $\mathcal{F} = \{\Omega, \emptyset\}$ as this forces $\mathbb{E}[X|\mathcal{F}] = \mathbb{E}[X]$ almost surely , $\mathbb{E}[\exp(\lambda X - \mathbb{E}[X|\mathcal{F}])|\mathcal{F}] = \mathbb{E}[\exp(\lambda X - \mathbb{E}[X])]$ almost surely, and $A, B$ to be constant almost surely.

*Proof.* We will just sketch the argument (since the proof of Hoeffding's lemma is complete). As in the unconditional case we have, by monotonicity of unconditional expectation, that

$$A \leq X - \mathbb{E}[X|\mathcal{F}] \leq B \text{ a.s.} \implies \mathbb{E}[A|\mathcal{F}] \leq \mathbb{E}[X - \mathbb{E}[X|\mathcal{F}]] \leq \mathbb{E}[B|\mathcal{F}] \text{ a.s} \implies A \leq 0 \leq B \text{ a.s.}$$

with the last inequality follow from assumption that $A$ and $B$ are $\mathcal{F}$ measurable.

We define the following events:

$$E_1 = \{A < B\},\ E_2 = \{A = B\},\ E_3 = \{A > B\},\ H = \{A = B = 0\}.$$

From the earlier argument we have that $1_{\mathbb{E}_3} = 1_{E_2 \cap H^c} = 0$ almost surely. Since all of these events are in $\mathcal{F}$, $\Omega = E_1 \cup E_2 \cup E_3$ and $H \subset E_2$ we have that

$$\mathbb{E}[\exp(\lambda(X - \mathbb{E}[X|\mathcal{F}]))|\mathcal{F}] = (\mathbf{1}_{E_1} + \mathbf{1}_H)\mathbb{E}[\exp(\lambda(X - \mathbb{E}[X|\mathcal{F}]))|\mathcal{F}] \text{ a.s.}$$

Now the condition $A \leq X - \mathbb{E}[X|\mathcal{F}] \leq B$ implies that $\lambda(X - \mathbb{E}[X|\mathcal{F}]) \leq \max(\lambda A, \lambda B)$, regardless of the choice of $\lambda$. Since $1_H \exp(\max(\lambda A, \lambda B)) = 1_H \cdot 1 \leq 1_H \exp(\lambda^2(B-A)^2/8)$ almost surely, monotonicity of conditional expectation implies that

$$1_H \mathbb{E}[\exp(\lambda(X - \mathbb{E}[X|\mathcal{F}]))|\mathcal{F}] \leq 1_H \exp(\lambda^2(B-A)^2/8) \text{ a.s.}$$

We now turn to bounding $\mathbf{1}_{E_1}\mathbb{E}[\exp(\lambda(X-\mathbb{E}[X|\mathcal{F}]))|\mathcal{F}]$. The convexity map argument from before implies that

$$\mathbf{1}_{E_1}\mathbb{E}[\exp(\lambda(X-\mathbb{E}[X|\mathcal{F}]))|\mathcal{F}] \leq \mathbf{1}_{E_1}\left(\frac{B}{B-A}\exp(\lambda A)+\frac{-A}{B-A}\exp(\lambda B)\right) \ a.s.,$$

where the RHS is understood to be 0 if $\mathbf{1}_{E_1}=0$. (10) then implies that

$$\frac{B}{B-A}\exp(\lambda A)+\frac{-A}{B-A}\exp(\lambda B) \leq \exp(\lambda^2(B-A)^2/8) \qquad \forall \omega \in E_1,$$

Therefore

$$\mathbf{1}_{E}\mathbb{E}[\exp(\lambda(X-\mathbb{E}[X|\mathcal{F}]))|\mathcal{F}] \leq \mathbf{1}_{E}\exp(\lambda^2(B-A)^2/8) \qquad \text{almost surely.}$$

The result follows. □

We now prove the concentration inequalities:

**Theorem 12** (Hoeffding's Inequality)**.** *Let $X_1,\cdots,X_n$ be independent random variables such that $a_i \leq X_i \leq b_i$ almost surely for all $1\leq i\leq n$. Denote $X=\sum_{i=1}^n X_i$. Then, for all $t\geq 0$.*

$$\mathbb{P}(|X-\mathbb{E}[X]|>t) < 2\exp\left(\frac{2t^2}{\sum_{i=1}^n(b_i-a_i)^2}\right).$$

*Proof.* We first address the upper tail case. Here we assume that $t\geq 0$. We let $\lambda$ denote a positive value that we'll specify later. Since $\lambda>0$ and the map $x\mapsto\exp(x)$ is monotone increasing we have that

$$\mathbb{P}(X-\mathbb{E}[X]>t)=\mathbb{P}(\exp(\lambda(X-\mathbb{E}[X]))>\exp(\lambda t)) < \exp(-\lambda t)\mathbb{E}\exp(\lambda(X-\mathbb{E}[X])),$$

with the last inequality being Markov. By multiplicativity of $\exp(\cdot)$, independence and Hoeffding's Lemma we have

$$\mathbb{E}\exp(\lambda(X-\mathbb{E}[X])) = \prod_{i=1}^n \mathbb{E}\exp(\lambda(X_i-\mathbb{E}[X_i])) \leq \prod_{i=1}^n \exp(\lambda^2(b_i-a_i)^2/8).$$

Letting $f(\lambda):=-\lambda t+\frac{\lambda^2}{8}\sum_{i=1}^n(b_i-a_i)^2$ one can check that $f'(\lambda)=0 \iff \lambda = 4t/\sum_{i=1}^n(b_i-a_i)^2$ and that $\inf_\lambda f''(\lambda)=\frac{1}{4}\sum_{i=1}^n(b_i-a_i)^2\geq 0$. In particular the extreme point is a minimizer. Therefore, taking $\lambda = 4t/\sum_{i=1}^n(b_i-a_i)^2$ gives

$$\mathbb{P}(X-\mathbb{E}[X]>t) < \exp(-\lambda t)\mathbb{E}\exp(\lambda(X-\mathbb{E}[X])) \leq \exp(f(\lambda)) = \exp\left(-\frac{2t^2}{\sum_{i=1}^n(b_i-a_i)^2}\right)$$

This deals with the upper tail. For the lower tail we assume that $t\leq 0$. Note that

$$\mathbb{P}(X-\mathbb{E}[X]<t)=\mathbb{P}(-X-\mathbb{E}[-X]>-t).$$

Since $-t\geq 0$ and $-b_i\leq -X\leq -a_i$ almost surely we may apply the uppertail argument here to conclude that

$$\mathbb{P}(X-\mathbb{E}[X]<t)=\mathbb{P}(-X-\mathbb{E}[-X]>-t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n(b_i-a_i)^2}\right).$$

Combining the tail bounds gives the desired result. □

A generalization of Hoeffding's inequality is the Azuma-Hoeffding inequality, giving the same type of concentration inequality but for a martingale sequence. In the statement below we assume a common probability space $(\mathbb{P}, \Omega, \Sigma)$ for all random variables that appear.

**Theorem 13** (Azuma-Hoeffding)**.** *Let $\mathcal{F} = (\mathcal{F}_n)_{n\geq 0}$ be a filtration w.r.t $\Sigma$. Let $(X_n)_{n\geq 0}$ be a martingale sequence that's adapted to $\mathcal{F}$ (i.e. each $X_n$ is integrable, $\mathcal{F}_n$-measurable and $\mathbb{E}[X_n|\mathcal{F}_m] = \mathbb{E}[X_m]$ whenever $m \leq n$). Let $(A_n)_{n\geq 1}, (B_n)_{n\geq 1}$ be an adapted processes with respect to $\mathcal{F}$ (i.e. both $A_n$ and $B_n$ are $\mathcal{F}_{n-1}$ measurable) that satisfies $A_n \leq X_n - X_{n-1} \leq B_n$ almost surely and $B_n - A_n \leq c_n$ almost surely, where $(c_n)_{n\geq 1}$ is a sequence of non-negative numbers. Then, for every $n \geq 1$.*

$$\mathbb{P}(|X_n - X_0| > t) < 2\exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right)$$

The Hoeffding's inequality is a corollary of the Azuma-Hoeffding inequality since, given the sequence of random variables $(Z_n)_{n\geq 1}$, with $a_n \leq Z_n \leq b_n$ almost surely, we can define the following martingale sequence $(X_n)_{n\geq 1}$:

- $X_0 = 0, X_{k+1} = \sum_{i=1}^{k+1} Z_{k+1} - \mathbb{E}[Z_{n+1}]$ for all $0 \leq k \leq n-1$.
- $\mathcal{F}_n = \sigma(X_1, \cdots, X_n)$ (the smallest sub-sigma algebra of $\Sigma$ for which $X_1, \cdots, X_n$ are measurable).

Then $(a_n)_{n\geq 1}, (b_n)_{n\geq 1}$ is adapted to $\mathcal{F}$ trivially and $X_n - X_0 = \sum_{i=1}^n Z_n - \mathbb{E}[Z_n] = Z - \mathbb{E}[Z]$.

*Proof.* We will follow the proof of Hoeffding's inequality. We first consider the upper-deviation case. For the proof of that case we will weaken the assumption on $(X_n)_{n\geq 0}$ to that of a supermatingale (so $\mathbb{E}[X_n|\mathcal{F}_m] \leq X_m$ for all $m \leq n$). By Doob's Martingale-Decomposition Theorem we can write $X_n = Y_n + Z_n$, where $Y_n$ is a martingale and $Z_n$ is a non-increasing adapted process w.r.t. $\mathcal{F}$.

As in the proof of Hoeffding's inequality the crux of the matter is bounding $\mathbb{E}\exp(\lambda(X_n - X))$, for some particular $\lambda > 0$. To that end we have that

$$\begin{aligned}
\mathbb{E}\exp(\lambda(X_n - X_0))] &= \mathbb{E}\prod_{i=1}^{n}\exp(\lambda(X_i - X_{i-1})) \\
&= \mathbb{E}\prod_{i=1}^{n-1}\exp(\lambda(X_i - X_{i-1}))\mathbb{E}[\exp(\lambda(X_n - X_{n-1}))|\mathcal{F}_{n-1}] \\
&\leq \mathbb{E}\prod_{i=1}^{n-1}\exp(\lambda(X_i - X_{i-1}))\mathbb{E}[\exp(\lambda(Y_n - \mathbb{E}[Y_n|\mathcal{F}_{n-1}]))|\mathcal{F}_{n-1}] \\
&\leq \mathbb{E}\prod_{i=1}^{n-1}\exp(\lambda(X_i - X_{i-1}))\exp(\lambda^2(B_n - A_n)^2/8) \\
&\leq \mathbb{E}\prod_{i=1}^{n-1}\exp(\lambda(X_i - X_{i-1}))\exp(\lambda^2 c_n^2).
\end{aligned}$$

For the first inequality we used the assumption that $Z_n - Z_{n-1} \leq 0$ almost surely (The $Z_n$ are non-increasing) and the fact that $Y_{n-1} = \mathbb{E}[Y_n|\mathcal{F}_{n-1}]$. In the second inequality we used the conditional Hoeffding's lemma. For the last inequality we used the assumption that $B_n - A_n \leq c_n$ almost surely. Iterating the conditional expectation argument for $\mathcal{F}_{n-2}, \cdots, \mathcal{F}_0$ gives

$$\mathbb{E}\exp(\lambda(X_n - X_0))] \leq \prod_{i=1}^{n}\exp(\lambda^2 c_n^2).$$

At this point the proof is exactlty the same as in Hoeffding's inequality for the upper tail-bound. For the lower tail bound we relax the assumption to the $X$'s forming a submartingale (so $\mathbb{E}[X_n|\mathcal{F}_m] \geq X_m$ whenever $m \leq n$). Since the negation of a submartingale sequence is a supermartingale sequence the same transfer argument of upper tail-bounds to lower-tail bounds from the proof of Hoeffding's inequality applies here as well. This gives the desired upper and lower tail bounds for $X_n - X_0$ and adding the two together gives the desired concentration inequality. □

We now prove McDiarmid's inequality.

**Theorem 14.** *Let $X_1, \cdots, X_n$ be a collection of independent random variables and write $X := (X_1, \cdots, X_n)$. Let $f : \mathbb{R}^n \to \mathbb{R}$ satisfy the condition*

$$\sup_{x \in \mathbb{R}^n} \sup_{t \in \mathbb{R}} |f(x) - f(x + te_i)| \leq c_i, \qquad \forall 1 \leq i \leq n$$

*where $e_i$ is the ith standard basis vector and $(c_i)_{i \geq 1}$ is a non-negative sequence of numbers. Then*

$$\mathbb{P}\left(|f(X) - \mathbb{E}f(X)| > t\right) < 2\exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right)$$

*Proof.* The proof is an application of Azuma-Hoeffding. We define the following martingale sequence $(Z_n)_{n \geq 1}$

1. $\mathcal{F}_0 = \{\Omega, \emptyset\},\ \mathcal{F}_n = \sigma(X_1, \cdots, X_n)$
2. $Z_n = f(X),\ Z_m = \mathbb{E}[Z_n|\mathcal{F}_m]$ for all $0 \leq m \leq n$.

Note that the construction is well-defined since $Z_n$ is integrable ($|Z_n| \leq \sup_{x \in \mathbb{R}^n} f(x) \leq |f(0)| + \sum_{i=1}^n c_i < \infty$ almost surely). This construction of a martingale sequence from a random variable and a filtration is commonly referred to as a Doob-Martingale. Since $Z_0 = \mathbb{E}[Z_n]$ the only thing left to check is the condition that $A_m \leq Z_m - Z_{m-1} \leq B_m$ for every $1 \leq m \leq n$, where $(A_m)_{m \geq 1}, (B_m)_{m \geq 1}$ is an adapted process w.r.t $\mathcal{F}$. Naively one could take $A_m = -c_m, B_m = c_m$ but that would lose a factor of 4 in the exponent. Instead, using the fact that $A_m, B_m$ must be $\mathcal{F}_{m-1}$ measurable, we will take

$$A_m = \mathbb{E}\left[\inf_{t \in \mathbb{R}} f(X + (t - X_m)e_m)|\mathcal{F}_m\right] - Z_{m-1}, \qquad B_m = \mathbb{E}\left[\sup_{t \in \mathbb{R}} f(X + (t - X_m)e_m)|\mathcal{F}_m\right] - Z_{m-1}.$$

Notice that $A_m$ and $B_m$ are actually $\mathcal{F}_{m-1}$ measurable since both $\mathbb{E}[\inf_{t \in \mathbb{R}} f(X + (t - X_m)e_m)|\mathcal{F}_m]$ and $\mathbb{E}[\sup_{t \in \mathbb{R}} f(X + (t - X_m)e_m)|\mathcal{F}_m]$ are independent of $X_m$. By monotonicity of conditional expectation

$$\begin{aligned}\mathbb{E}\left[\inf_{t \in \mathbb{R}} f(X + (t - X_m)e_m) \mid \mathcal{F}_m\right] &\leq \mathbb{E}[f(X) \mid \mathcal{F}_m] \\ &\leq \mathbb{E}\left[\sup_{t \in \mathbb{R}} f(X + (t - X_m)e_m) \mid \mathcal{F}_m\right] \quad \text{almost surely.}\end{aligned}$$

In addition,

$$\begin{aligned}B_m - A_m &= \mathbb{E}\left[\sup_{t \in \mathbb{R}} f(X + (t - X_m)e_m)|\mathcal{F}_m\right] - \mathbb{E}\left[\inf_{t \in \mathbb{R}} f(X + (t - X_m)e_m)|\mathcal{F}_m\right] \ a.s. \\ &\leq \mathbb{E}\left[\sup_{s,t \in \mathbb{R}} |f(X + (t - X_m)e_m) - f(X + (s - X_m)e_m)||\mathcal{F}_m\right] \ a.s. \\ &\leq c_m \qquad \text{almost surely,}\end{aligned}$$

The first inequality follows from conditional linearity of expectation.The last inequality follows from the bounded difference assumption on $f$ and the monotonicity of conditional expectation. Thus we may apply Azuma-Hoeffding to the martingale sequence and conclude the desired concentration inequality for $|Z_n - Z_0|$. Since $Z_n - Z_0 = f(X) - \mathbb{E}[f(X)]$ the result follows. □

Lastly, we will give a nice application of concentration inequalities from Random Graph theory. For the Erdos-Renyi model $G_{n,p}$, a random graph $G \sim G_{n,p}$ is supported on the set of all graphs on $n$ vertices and, for any fixed graph $H$, satisfies

$$\mathbb{P}(G = H) = p^{|E(H)|}(1-p)^{\binom{n}{2}-|E(H)|}$$

Because of the law of $G_{n,p}$, $G$ itself can be identified with a random indicator vector in $\{0,1\}^{\binom{n}{2}}$ whose entries are distributed as independent Bernoulli $p$-random variables (with each entry in the vector and each pair of vertices in $G$ being in 1-to-1 correspondence). For a graph $H$ the chromatic number $\chi(H)$ is the fewest number of colors needed to color all vertices of $H$ without $H$ having an adjacent pair of same-colored vertices.

**Corollary 3** (Chromatic number of random graphs)**.** *If $G \sim G_{n,p}$ then*

$$\mathbb{P}(|G - \mathbb{E}[\chi(G)]| > t) < 2\exp\left(-\frac{t^2}{2n}\right)$$

*Proof.* As mentioned earlier we may view $G$ as a random indicator vector where each entry is distributed as a $p$-Bernoulli. In this way $\chi(G)$ can be viewed as a function of independent bounded random variables. Since changing the edge-status of a single vertex pair only changes the chromatic number by 1, a naive application of McDiarmid's inequality gives

$$\mathbb{P}(|\xi(G) - \mathbb{E}[\chi(G)]| > t) < 2\exp\left(-\frac{2t^2}{\binom{n}{2}}\right).$$

This, however, is essentially useless since the RHS is bigger than 1 until $t = n$ and the chromatic number of every graph on $n$ vertices is at most $n$. Instead, one can take an appropriate Doob-Martingale and apply Azuma-Hoeffding. The naive approach would be to generate the Doob Martingale from the finest filtration (take an arbitrary ordering of the vertex pairs and take $\mathcal{F}_k$ to be the sub-sigma algebra generated by the first $k$ of them). That of course is exactly the same thing as applying McDiarmid's inequality. Instead, one can take a coarser filtration. In particular one can order the vertices of $G$ (into say $v_1, v_2, \cdots$) and partition all vertex pairs into sets $S_1, \cdots, S_{n-1}$, where $S_i$ consists of all vertex pairs where the the first vertex in the ordering to appear in the pair is $v_i$. Then one can define $\mathcal{F}_k$ to be the sub-sigma algebra generated by the random variables indexed by $S_1 \cup \cdots \cup S_k$. Taking $(Z_k)_{k\geq 0}$ to be the associated Doob-Martingale sequence, the point is that $|Z_k - Z_{k-1}| \leq 1$ almost surely for all $k \geq 1$. This is because any proper-coloring of $G$ can be modified to be a proper coloring of $\tilde{G}$ (where $\tilde{G}$ is any graph whose edges agree with $G$ on all vertex pairs not contained in $S_k$) by changing the color of $v_k$. Therefore Azuma-Hoeffding gives

$$\mathbb{P}(|Z_n - Z_0| > t) < 2\exp\left(-\frac{2t^2}{n}\right).$$

Since $Z_n = \chi(G), Z_0 = \mathbb{E}[\chi(G)]$ the result follows. □

# 9 Subgaussian and subexponential random variables

In the previous section there were two recurring components in our arguments for proving concentration inequalities. The first was using Markov's inequality to argue that a

$$\mathbb{P}(X - \mathbb{E}X > t) < \exp(-\lambda t)\mathbb{E}\exp(\lambda(X - \mathbb{E}X)),$$

where $\lambda$ is some non-negative parameter that is specified at the end of the argument. The second component was obtaining the bound

$$\mathbb{E}\exp(\lambda(X - \mathbb{E}X)) \leq \exp(\lambda^2 K^2),$$

Where $K$ is some parameter that depends only on $X$. In Hoeffding's inequality and its other variations this $K$ was equal to $\sum_{i=1}^n (b_i - a_i)^2/8$, where $a_i, b_i$ were almost surely bounds on $X_i - \mathbb{E}X_i$ where $X = X_1 + \cdots + X_n$ and the $X_i$ were independent. As it turns out this second component is a general property that a large class of random variables satisfy, called "subgaussian" random variables. Unsurprisingly, such random variables have similar concentration propertis to that of sums of independent bounded random variables (like in Hoeffding's). However, it turns out that their type of tail bound and the second component are equivalent characterization of subgaussian random variables. In fact there are a couple of other equivalent characterizations as well.

**Theorem 15** (Equivalent subgaussian properties)**.** *For a real random variable $X$, the following properties are equivalent, with the parameters $K_i > 0$ differing by at most an absolute constant factor:*

1. $\mathbb{P}\{|X| \geq t\} \leq 2e^{-t^2/K_1^2}$ *for all* $t \geq 0$.
2. $\|X\|_{L^p} \leq K_2\sqrt{p}$ *for all* $p \geq 1$.
3. $\mathbb{E}\exp(X^2/K_3^2) \leq 2$.
4. *If $X$ has mean 0 then.*
$$\mathbb{E}e^{\lambda X} \leq e^{K_4^2\lambda^2} \qquad \text{for every } \lambda \in \mathbb{R}.$$

*Proof.* We will show that $1 \implies 2 \implies 3 \implies 4 \implies 1$ and $3 \implies 1$. For convenience, when proving an implication we will assume that the premise holds for some parameter $K$ and prove that the conclusion holds for the parameter $CK$ where $C > 0$ is some absolute constant.

- $1 \implies 2$: The bath-tub formula for non-negative random variables and Fubini-Tonelli gives
$$\begin{aligned}\mathbb{E}|X|^p &= \int_0^\infty \mathbb{P}\{|X|^p > u\}\,du = \int_0^\infty pt^{p-1}\mathbb{P}\{|X| > t\}\,dt \\ &\leq 2p\int_0^\infty t^{p-1}e^{-t^2/K^2}\,dt = pK^p\int_0^\infty v^{p/2-1}e^{-v}\,dv = pK^p\Gamma(p/2).\end{aligned}$$
The substitutions are $u = t^p$ and then $v = t^2/K^2$. Here $\Gamma(s) = \int_0^\infty v^{s-1}e^{-v}\,dv$ is the gamma function. The estimate $\Gamma(s) \leq 3s^s$ for $s \geq 1/2$ implies
$$\|X\|_{L^p} \leq K\big(3p(p/2)^{p/2}\big)^{1/p} \leq CK\sqrt{p},$$
since $(3p)^{1/p}$ is bounded for $p \geq 1$.

- $2 \implies 3$: Because the terms appearing in the Taylor expansion of $x \mapsto \exp(x^2/K^2)$ are non-negative, the monotone convergence theorem allows us to pass the expectation inside the series expansion of $\exp(X^2/K^2)$ when computing its expectation. In particular we get that

$$\begin{aligned}\mathbb{E}e^{X^2/(CK)^2} &= 1 + \sum_{j=1}^{\infty} \frac{\mathbb{E}X^{2j}}{j!(CK)^{2j}} \\ &\leq 1 + \sum_{j=1}^{\infty} \frac{K^{2j}(2j)^j}{(j/e)^j(CK)^{2j}} = \sum_{j=0}^{\infty} \left(\frac{2e}{C}\right)^j.\end{aligned}$$

  For the inequality we used (2) and the fact that $j! \geq (j/e)^j$ for all $j \geq 0$. Note now that the geometric series is at most 2 when $C \geq 4e$.

- $3 \implies 4$: The proof of this is the most involved. Take $Y = X/K$. Then by (3) and the mean 0 assumption $\mathbb{E}Y = 0$ and $\mathbb{E}\exp(Y^2) \leq 2$. Furthermore by homogeneity it suffices to prove (4) with $Y$ in place of $X$ and $C$ in place of $CK$. We will need the following consequence of Taylor's theorem:

$$e^u \leq 1 + u + u^2 e^{|u|}/2 \qquad \forall u \in \mathbb{R}.$$

  In addition, we will need the following inequality:

$$y^2 e^{|\lambda y|} \leq e^{\lambda^2/2} e^{y^2} \qquad \forall y, \lambda \in \mathbb{R}.$$

  This is basically a consequence of the facts that $y^2 \leq e^{y^2/2}$ (Since $e^{y^2/2} - y^2 \geq e^{-y^2/2} \geq 0$) and that $|\lambda y| \leq (\lambda^2 + y^2)/2$. Thus we can apply the consequence of Taylor's theorem and the other facts to get

$$\begin{aligned}\mathbb{E}\exp(\lambda Y) &\leq 1 + \mathbb{E}[\lambda Y] + \mathbb{E}[\lambda^2 Y^2 \exp(|\lambda Y|)/2] \\ &\leq 1 + (\lambda^2/2)e^{\lambda^2/2}\mathbb{E}e^{Y^2} \\ &\leq 1 + \lambda^2 e^{\lambda^2/2} \\ &\leq (1+\lambda)^2 e^{\lambda^2/2} \leq e^{3\lambda^2/2}.\end{aligned}$$

  In particular can obtain the conclusion with $C = 3/2$.

- $4 \implies 1$: This is exactly the same argument as in the proof of Hoeffding's inequality, with (4) being used in place of Hoeffding's inequality. Using the same Markov approach with (4) gives

$$\mathbb{P}(|X| > t) < 2\exp(-t^2/(4K^2)).$$

- $3 \implies 1$: This proof is the easiest. Markov's inequality gives

$$\mathbb{P}(|X| > t) = \mathbb{P}(X^2/K^2 > t^2/K^2) < \exp(-t^2/K^2)\mathbb{E}\exp(X^2/K^2) \leq 2\exp(-t^2/K^2).$$

□

As a comment in the proof of $3 \implies 1$ the exponent of $-t^2/4K^2$ is exactly we would expect (in view of Hoeffding's inequality) since there the exponent is $-t^2/\sum_i (b_i - a_i)^2$ and $K = \sum_i (b_i - a_i)^2/8$.

We now focus on the 3rd characterization of a subgaussian random variable. It will be convenient to compare subgaussian random variables in terms of this characterization.

**Definition 12** (Subgaussian norm)**.** A random variable is *subgaussian* if it satisfies the equivalent properties in Theorem 15. Its subgaussian norm is

$$\|X\|_{\psi_2} = \inf\{K > 0 : \mathbb{E}\exp(X^2/K^2) \le 2\}.$$

These are Orlicz norms: the functions defining them will be $u \mapsto e^{u^2} - 1$ and $u \mapsto e^u - 1$ for $u \ge 0$. Bounded random variables are subgaussian, since $|X| \le B$ implies $\|X\|_{\psi_2} \le B/\sqrt{\log 2}$. For $g \sim N(0, \sigma^2)$, direct Gaussian integration gives $\mathbb{E}e^{g^2/K^2} = (1 - 2\sigma^2/K^2)^{-1/2}$ when $K^2 > 2\sigma^2$; thus $\|g\|_{\psi_2} = \sqrt{8/3}\,\sigma$.

A nice fact about subgaussian norms is that $|\mathbb{E}X| \le \mathbb{E}|X| \le C\|X\|_{\psi_2}$. This last inequality is because Jensen's inequality implies that

$$\exp((\mathbb{E}|X|)^2/\|X\|_{\psi_2}^2) \le \mathbb{E}\exp(X^2/\|X\|_{\psi_2}^2) \le 2 \implies \mathbb{E}|X| \le \|X\|_{\psi_2}\sqrt{\log 2}$$

Therefore, because $\|\cdot\|_{\psi_2}$ is a norm, we get

$$\|X - \mathbb{E}X\|_{\psi_2} \le \|X\|_{\psi_2} + \|\mathbb{E}X\|_{\psi_2} \le C\|X\|_{\psi_2}.$$

A very nice fact about the subgaussian norm is the following "triangle inequality" that is satisfies. This essentially allows one to conclude that sums of independent subgaussian random variables are subgaussian.

**Theorem 16** (Subgaussian sums)**.** *Let $X_1, \dots, X_n$ be independent, mean-zero, subgaussian random variables. Then*

$$\left\|\sum_{i=1}^n X_i\right\|_{\psi_2}^2 \le C\sum_{i=1}^n \|X_i\|_{\psi_2}^2, \qquad \mathbb{P}\left\{\left|\sum_{i=1}^n X_i\right| \ge t\right\} \le 2\exp\left(-\frac{ct^2}{\sum_i \|X_i\|_{\psi_2}^2}\right)$$

*for every $t \ge 0$.*

*Proof.* First note that if $\|X_i\|_{\psi_2} = 0$ then $X_i = 0$ almost surely (recall that Jensen implies that $\mathbb{E}X_i^2 \le \|X_i\|_{\psi_2}$ and $\mathbb{E}X_i^2 = 0 \implies X_i = 0$ almost surely). Thus we may assume that all the $X_i$ have non-zero $\psi_2$ norm. Next, since the $X_i$ have mean 0 and are independet we may use Hoeffding-type characterization of subgaussian random variables to conclude that

$$\mathbb{E}\exp\left(\lambda\sum_{i=1}^n X_i\right) = \prod_{i=1}^n \mathbb{E}\exp\lambda(X_i - \mathbb{E}X_i) \le \prod_{i=1}^n \exp(C\lambda^2\|X_i\|_{\psi_2}^2) = \exp\left(\lambda^2 C\sum_{i=1}^n \|X_i\|_{\psi_2}^2\right),$$

where $C > 0$ is some absolute constant. Therefore the third characterization of subgaussian random variables gives gives the first conclusion. The second conclusion follows from the equivalence of the first and third characterizations of subgaussian random variables. □

In particular, if $\|X_i\|_{\psi_2} \le K$ and $a \in \mathbb{R}^n$, then the same argument applied to $a_iX_i$ bounds the norm of $\sum_i a_iX_i$ by $CK\|a\|_2$.

A closely related class of random variables are subexponential random variables. These are random variables for which $-\log(\mathbb{P}(|X| > t))$ is linear in $t$. This is in contrast to subgaussian random variables, where $-\log(\mathbb{P}(|X| > t))$ is quadratic in $t$. Just like subgaussian random variables, exponential random variables have an equivalent set of characterizations.

**Theorem 17** (Equivalent subexponential properties)**.** *For a real random variable $X$, the following properties are equivalent, with the parameters differing by at most an absolute constant factor:*

1. $\mathbb{P}\{|X| \geq t\} \leq 2e^{-t/K_1}$ *for all* $t \geq 0$.
2. $\|X\|_{L^p} \leq K_2 p$ *for all* $p \geq 1$.
3. $\mathbb{E}\exp(|X|/K_3) \leq 2$. *If* $\mathbb{E}X = 0$, *these are also equivalent to*

$$\mathbb{E}e^{\lambda X} \leq e^{K_4^2 \lambda^2} \qquad \text{whenever } |\lambda| \leq K_4^{-1}.$$

The proof of the equivalence of the characterizations of the subexponential random variable is almost identical to the proof for subgaussian random variables. We will however prove the implication (3) $\implies$ (4) since the proof is somewhat different (although the argument is similar).

As in the subgaussian case, by homogeneity it suffices to assume that $K_3 = 1$ and prove (4) with $K_4$ an absolute constant. We will need the following facts:

- $e^u \leq 1 + u + \frac{u^2}{2}\exp(|u|)$ for all $u \in \mathbb{R}$.
- $u^2/4 \leq e^{|u|/2}$ for all $u \in \mathbb{R}$.

The first fact can be seen by observing that, for $u \neq 0$

$$\frac{e^u - 1 - u}{u^2/2} = \sum_{n\geq 2} \frac{u^{n-2}}{n!/2} = \sum_{n\geq 0} \frac{u^n}{(n+2)!/2} \leq \sum_{n\geq 0} \frac{|u|^n}{n!} = e^{|u|}.$$

The second fact can be seen by observing that, for $u \neq 0$,

$$e^{|u|/2} - \frac{u^2}{4} = 1 + \frac{|u|}{2} - \frac{|u|^2}{8} + \sum_{n\geq 3} \frac{|u|^n}{2^n n!} \geq 2 - 1 + \frac{|u|}{2} - \frac{|u|^2}{8} - \sum_{n\geq 3} \frac{((-1)|u|)^n}{2^n n!} = 2 - \exp(-|u|/2) \geq 1$$

Returning to the problem, these facts imply that

$$\begin{aligned}
\mathbb{E}\exp(\lambda X) &\leq 1 + \mathbb{E}\left[\frac{\lambda^2 X^2}{2}\exp(|\lambda X|)\right] \\
&\leq 1 + \mathbb{E}\left[2\lambda^2 \exp(|\lambda X| + |X|/2)\right] \\
&\leq 1 + \mathbb{E}[2\lambda^2 \exp(|Y|)] \\
&\leq 1 + 4\lambda^2 \leq e^{4\lambda^2}.
\end{aligned}$$

For the third inequality we only required that $\lambda \leq 1/2$. Given that and the last inequality we can conclude (4) with $K = 2$.

As in the subgaussian case, there is also a notion of a subexponential norm.

**Definition 13** (Subexponential norm)**.** A variable satisfying these equivalent properties is *subexponential*, with norm

$$\|X\|_{\psi_1} = \inf\{K > 0 : \mathbb{E}\exp(|X|/K) \leq 2\}.$$

**Lemma 10** (Squares and products)**.** *$X$ is subgaussian if and only if $X^2$ is subexponential, and*

$$\|X^2\|_{\psi_1} = \|X\|_{\psi_2}^2.$$

*If $X, Y$ are subgaussian, without any independence assumption, then*

$$\|XY\|_{\psi_1} \leq \|X\|_{\psi_2}\|Y\|_{\psi_2}.$$

*Proof.* For the first conclusion observe that $\mathbb{E}\exp(X^2/L^2) = \mathbb{E}\exp(|X^2|/L^2)$ for any positive parameter $L > 0$. Therefore the definition of $\|X^2\|_{\psi_1}$ is equivalent to that of $\|X\|_{\psi_2}$ and so the equality of the values is automatic. For the second conclusion note that the simple algebraic identity $xy \leq (x^2 + y^2)/2$ implies that any two random variables satisfy $XY \leq X^2/2 + Y^2/2$ almost surely. Therefore

$$\mathbb{E}\exp(XY) \leq \mathbb{E}\exp(X^2/2 + Y^2/2) \leq \sqrt{\mathbb{E}\exp(X^2)\mathbb{E}\exp(Y^2)}.$$

Here the last second to last inequality is just Holder's inequality. Replacing $X$ and $Y$ with $X/K_1$ and $Y/K_2$ with $K_1 > \|X\|_{\psi_2}, K_2 > \|Y\|_{\psi_2}$, by homogeneity the right most inequality is at most 2. Thus $\|XY\|_{\psi_1} \leq K_1 K_2$. Taking taking $K_1 \downarrow \|X\|_{\psi_2}, K_2 \downarrow \|Y\|_{\psi_2}$ we get the second conclusion.

□

Centering also satisfies $\|X - \mathbb{E}X\|_{\psi_1} \leq C\|X\|_{\psi_1}$, by the same triangle-inequality argument. Thus a centered square has subexponential norm at most $C\|X\|_{\psi_2}^2$.

**Theorem 18** (Subexponential Bernstein inequality)**.** *Let $X_1, \ldots, X_n$ be independent, mean-zero, subexponential random variables. For every $t \geq 0$,*

$$\mathbb{P}\left\{\left|\sum_{i=1}^n X_i\right| \geq t\right\} \leq 2\exp\left[-c\min\left\{\frac{t^2}{\sum_i \|X_i\|_{\psi_1}^2}, \frac{t}{\max_i \|X_i\|_{\psi_1}}\right\}\right].$$

*Proof.* Let $S = \sum_{i=1}^n X_i$, $K_i = \|X_i\|_{\psi_1}$, $V = \sum_i K_i^2$, and $B = \max_i K_i$. If $K_i = 0$ then $X_i = 0$ almost surely. Since the inequality is trivial when the sum of the $X_i$ is 0 almost surely, we may assume that $K_i > 0$ for $i$. In addition, by homogeneity of the subexponential norm and form of the exponent in the tail bound we may assume that $\max_i \|X_i\|_{\psi_1} \leq 1$.

To apply characterization (4) of subexponential random variable to the Markov approach we will use the following observation: for $K > 0, t \geq 0$ consider the function

$$f(\lambda) := K^2\lambda^2 - \lambda t = K^2\left(\lambda - \frac{t}{2K^2}\right)^2 - \frac{t^2}{4K^2}.$$

Clearly the minimum is obtained at $\lambda_0 = t/2K^2$ with $f(\lambda_0) = -t^2/4K^2$. Now consider the map

$$h(\lambda) := \frac{f(\lambda_0) - f(0)}{\lambda_0 - 0}\lambda = -\lambda t/2.$$

Because $f$ is convex and $h$ is linear and $f(0) = h(0)$ and $f(\lambda_0) = h(\lambda_0)$ we have that $h(\lambda) \geq f(\lambda)$ on the interval $[0, \lambda_0]$.

Next, using the subgaussian triangle inequality we have the following subexponential triangle inequality

$$\left\|\sum_{i=1}^n X_i\right\|_{\psi_1} = \left\|\sqrt{\sum_{i=1}^n X_i}\right\|_{\psi_2}^2 \leq \left\|\sum_{i=1}^n \sqrt{|X_i|}\right\|_{\psi_2}^2 \leq C\sum_{i=1}^n \|\sqrt{|X_i|}\|_{\psi_2}^2 = C\sum_{i=1}^n \|X_i\|_{\psi_1}.$$

The subgaussian triangle inequality was used for the second inequality, the relation $\|X^2\|_{\psi_1} = \|X\|_{\psi_2}^2$ was used for the two equalities, and the first inequality follows from concavity of the square root function and the last inequality follows. In addition $\sum_{i=1}^n \|X_i\|_{\psi_1} \geq \sum_{i=1}^n \|X_i\|_{\psi_1}^2$ since $\max_i \|X_i\|_{\psi_1} = 1$. Therefore the subexponential norm characterization of exponential random variables implies that

$$\mathbb{E}\exp(\lambda S) \leq \exp(C_2\lambda^2 V) \qquad (|\lambda| \leq 1/C_2)$$

The local MGF characterization in the subexponential equivalence theorem provides absolute constants $C_0, c_0 > 0$ such that

$$\mathbb{E}e^{\lambda X_i} \le e^{C_0 \lambda^2 K_i^2} \qquad \text{if } |\lambda| \le c_0/K_i.$$

Indeed, its parameter $K_4$ can be chosen at most an absolute constant times $K_i$; increasing the coefficient in the exponent and decreasing the allowed interval makes these constants uniform in $i$. Since $K_i \le B$, the common restriction $|\lambda| \le c_0/B$ suffices for every factor. Independence then yields

$$\mathbb{E}e^{\lambda S} = \prod_{i=1}^{n} \mathbb{E}e^{\lambda X_i} \le \prod_{i=1}^{n} e^{C_0 \lambda^2 K_i^2} = e^{C_0 \lambda^2 V} \qquad (|\lambda| \le c_0/B).$$

For $t > 0$, exponential Markov therefore gives

$$\mathbb{P}\{S \ge t\} \le \exp(-\lambda t + C_0 \lambda^2 V) \qquad (0 < \lambda \le c_0/B).$$

The unconstrained minimum of this quadratic exponent occurs at $t/(2C_0V)$. Following the book, choose

$$\lambda_* = \min\left\{ \frac{t}{2C_0 V}, \frac{c_0}{B} \right\}.$$

This choice is admissible and satisfies $C_0 \lambda_* V \le t/2$. Consequently,

$$\begin{aligned} -\lambda_* t + C_0 \lambda_*^2 V &= -\lambda_* (t - C_0 \lambda_* V) \le -\frac{\lambda_* t}{2} \\ &= -\min\left\{ \frac{t^2}{4C_0 V}, \frac{c_0 t}{2B} \right\}. \end{aligned}$$

Writing $c = \min\{1/(4C_0), c_0/2\}$, we conclude that

$$\mathbb{P}\{S \ge t\} \le \exp\left[ -c \min\left\{ \frac{t^2}{V}, \frac{t}{B} \right\} \right].$$

The variables $-X_i$ remain independent and centered, with the same subexponential norms. Repeating the upper-tail argument for their sum bounds $\mathbb{P}\{S \le -t\}$ by the same expression. Adding the two bounds proves the assertion for $t > 0$, and $t = 0$ is trivial. The quadratic regime arises while the unconstrained choice is admissible; the linear regime arises once the MGF restriction forces the choice $\lambda_* = c_0/B$. □

For a weighted sum with $\|X_i\|_{\psi_1} \le K$, this becomes

$$\mathbb{P}\left\{ \left| \sum_i a_i X_i \right| \ge t \right\} \le 2 \exp\left[ -c \min\left\{ \frac{t^2}{K^2 \|a\|_2^2}, \frac{t}{K \|a\|_\infty} \right\} \right].$$

For the average of $n$ variables the exponent is $-cn \min\{t^2/K^2, t/K\}$ at deviation level $t$. Small deviations have a quadratic exponent; large deviations have a linear one.

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